Assignment 2

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1 The best classification model

The model for predicting High Average Score (HAS $= 1$ if score > 4.9) is constructed using a binomial family with a logit link function. Variable selection was initially performed using LASSO, yielding an accuracy of 0.7024, an AUC of 0.738, and an AIC of 7063.4. The regularization parameter $\lambda = 0.001979$ was selected based on the minimum binomial deviance rather than the more parsimonious one-standard-error choice, which provides the best prediction. However, our objective is not solely to maximize prediction accuracy, but to identify a model that provides richer information and more actionable insights. This approach offers greater flexibility in subsequent filtering(retaining more significant variables). Subsequently, we retained variables with coefficients significant at the 0.05 level. To improve interpretability and offer more insights for potential investors, we also included several informative variables that we are interested in. Finally, multicollinearity was assessed using VIF. This systematic approach strikes a balance between predictive performance and interpretability, as reflected in the final model's metrics: AUC = 0.725, accuracy = 0.702, AIC = 7107.4, along with acceptable VIF values(See appendix for the ROC curve and VIF results.) Although the predictive model performs worse than just choosing by LASSO, this gives us more useful information.

Here are the final results based on the full selection process. New variables were created through transformation. DistanceGroup was derived based on the distance to the city center (latitude -37.8124, longitude 144.9623), categorized into four groups: less than 2 km, 2–5 km, 5–10 km, and more than 10 km. Log(Price + 1) was also included as a transformation to address skewness. An interaction term, $RoomType * InstantBookable$, was intuitively created to explore potential combined effects. Specifically, we only present the variables that are significant at the 0.05 level for interpretability. For the full predictive model, please refer to the R code in the appendix.

$$
\begin{aligned}
\text{logit}(\mathbb{P}(\text{HAS}=1)) = & -2.934 - 0.4278\cdot\texttt{IdentityVerified} - 0.0099\cdot\texttt{AcceptanceRate} \\
& - 0.4567\cdot\texttt{HasProfilePic} + 0.0072\cdot\texttt{ResponseRate} + 0.8967\cdot\texttt{RoomTypePrivate} \\
& + 1.0150\cdot\texttt{IsSuperhost} - 0.0005\cdot\texttt{Availability365} - 0.0055\cdot\texttt{TotalListings} \\
& + 0.5474\cdot\texttt{Log(Price+1)} - 0.0002\cdot\texttt{MaxNightsAvgNTM} - 0.1794\cdot\texttt{InstantBookable} \\
& + 0.4096\cdot\texttt{VerificationEmailPhone} + 1.5280\cdot\texttt{Cardinia} + 0.7584\cdot\texttt{Hume} \\
& + 0.6365\cdot\texttt{YarraRanges} - 0.4841\cdot\texttt{DistanceGroup10kmPlus} - 0.0161\cdot\texttt{ReviewsLTM} \\
& + 0.1609\cdot\texttt{ReviewsL30D} - 0.4770\cdot\texttt{RoomPrivate\_InstantBookable} + \ldots
\end{aligned}
$$

Remark: One-hot encoding is applied to the categorical variables. *Cardinia*, *Hume*, and *Yarra Ranges* refer to neighbourhood names. Predictors that are not statistically significant at the 0.05 level are denoted as "…" in the table, with full details provided in the **Appendix**. The remaining variables are self-explanatory. During data cleaning, the variables `amenities`, `bathrooms text`, `has availability`, `property type`, `first review`, and `last review` are removed due to redundancy, irrelevance, or high processing cost. The `host since` variable is converted into the number of years the host has been active.

Interpreting the Logistic Model In this logistic model, qualitative predictors (e.g., `IsSuperhost`, `IdentityVerified`) are binary variables. A positive coefficient β_i implies that when the variable is "YES", the odds of $\mathbb{P}(\text{HAS}=1)$ increase by a factor of e^{β_i}. For instance, `IsSuperhost` with $\beta = 1.015$ increases the odds by $e^{1.015} \approx 2.76$. Conversely, a negative coefficient (e.g., `IdentityVerified` with $\beta = -0.428$) reduces the odds by a factor of $e^{-0.428} \approx 0.65$.

For quantitative predictors (e.g., `AcceptanceRate`, `Availability365`), a one-unit increase changes the log-odds by β_i. If $\beta_i < 0$, as in `AcceptanceRate` with $\beta = -0.0099$, each unit increase slightly decreases the odds (by a factor of $e^{-0.0099} \approx 0.9901$), thus reducing $\mathbb{P}(\text{HAS} = 1)$. In summary, positive coefficients increase the likelihood of HAS = 1, while negative ones reduce it.

2 How good is the logistic model?

As previously discussed, the logistic regression model strikes a practical balance between predictive accuracy and interpretability, achieving an AUC of 0.725, an accuracy of 0.702, and an AIC of 7107.4—all of which indicate a reasonably good fit for a binary classification task. Model adequacy was further assessed using the **residual plot(see appendix)**. The absence of strong patterns or curvature in the deviance residuals indicates no major violations of model assumptions, reinforcing the structural validity of the logistic regression.

To evaluate diagnostic soundness, we examined Variance Inflation Factors (VIFs) and found all values within acceptable thresholds(VIFs$<$ 2), suggesting no serious multicollinearity among predictors. A 10-fold cross-validation on the full dataset yielded a stable and low test MSE of approximately 0.192, supporting the model's generalizability. While the final model, built on manually selected significant predictors, performs well, it is slightly outperformed by the LASSO-regularized model (test MSE = 0.1897). This illustrates a common trade-off: enhancing interpretability by including more explanatory variables may slightly reduce predictive precision.

Ultimately, this trade-off is deliberate. The final model retains variables that are both statistically significant and practically relevant, offering richer interpretability and insights for decision-makers. While the LASSO-selected model may be more predictive, it omits variables of interest. Thus, our final model prioritizes interpretability without substantially sacrificing predictive performance.

3 Investment insights for improving HAS

We focus on features that new investors can directly influence—such as choosing suitable properties (e.g., location, property type, bathrooms, and beds) and managing their listings effectively (e.g., maintaining high response rates, proper verification, competitive pricing, etc.).

Starting with variables that have high coefficients ($|\beta_i| > 0.5$), owning a property in `Cardinia` has the strongest impact, increasing the odds of achieving HAS by $e^{1.5280} \approx 4.61$. Being a `Superhost` also significantly boosts the odds ($e^{1.015} \approx 2.76$). The variable `RoomTypePrivate` has a strong positive effect on HAS, increasing the log-odds by 0.8967. However, this effect is partially offset (-0.477) when the listing is also `InstantBookable`, as shown by the interaction term. Other beneficial factors include being located in `Hume` and `Yarra Ranges`, and offering a reasonable price.

Location Strategy: If budget allows, consider investing in properties located in `Cardinia`, `Hume`, or `Yarra Ranges`, and provide a `Private Room` listing. Avoid properties located more than 10km away from the CBD, as greater distance is associated with lower HAS probability. Surprisingly, locations within 10km do not show significant improvement, so for investors with tighter budgets, proximity to the CBD is less crucial than other factors.

Behavioral Strategy: For existing Airbnb hosts not currently achieving HAS, strive to become a `Superhost`, verify your profile with both `Email` and `Phone`, and encourage satisfied customers to leave reviews. Notably, reviews from the last 30 days have a tenfold stronger positive effect compared to the negative effect of older reviews, emphasizing the importance of recent feedback.

Maintaining a high `Response Rate` also positively affects guest satisfaction and review scores. Be responsive and professional. However, avoid accepting every booking—if you suspect a guest may leave a poor review based on bad behavior, it may be better to decline. Profile photos should reflect your listing accurately to avoid setting unrealistic expectations that could lead to low ratings.Finally, although offering a `Private Room` improves HAS probability, enabling `Instant Booking` may slightly reduce this advantage. Consider disabling this feature unless it is essential for your listing strategy.

4 The best Price Model

The best price model we chose was generated using **XGBoost**. The test MSE, RMSE, and R^2 were **6905.83**, **83.10**, and **0.695**, respectively, indicating superior performance over other methods.The **variable importance plot (see Appendix)** reveals that *room_typePrivate room* is by far the most influential predictor, followed by key features such as *bedrooms*, *bathrooms*, *accommodates*, and *distance_meters*. These suggest that room type, capacity, and location are central drivers of Airbnb pricing. Other notable predictors include *reviews_per_month*, *availability_365*, and host-related attributes like *host_listings_count* and *host_since_years*, indicating that both listing characteristics and host experience significantly influence price.

We applied the same data cleaning steps as in the training set. The distance to Melbourne CBD was calculated using geographic coordinates, and host experience was measured by calculating the difference between the listing's host date and the reference date (2025/07/29). Interaction terms and variable transformations were kept consistent with the training process. The data was split into 80% training and 20% test sets. **Why this is the best model:** We selected the **XGBoost** model for predicting price because it achieved the lowest test MSE and the highest test R^2, while maintaining acceptable generalization. We initially compared five models: LASSO, Stepwise Linear Regression, Ridge Regression, Random Forest, and XGBoost. Their performances are summarized below:

Model	**Test MSE**	**Test RMSE**	**Test R^2**	**Train R^2**
LASSO	10466.10	102.30	0.538	0.513
Linear Regression	10490.03	102.42	0.536	0.515
Random Forest	8626.71	92.88	0.619	0.889
Ridge	10419.71	102.08	0.540	0.515
XGBoost	**6905.83**	**83.10**	**0.695**	**0.892**

Table 1: Model performance comparison for Airbnb price prediction

Overall, the **XGBoost** model presents the lowest test MSE and the highest test and train R^2, indicating strong predictive accuracy and generalization ability.

In contrast, the linear models—LASSO, Ridge, and Linear Regression—produced very similar results, with test R^2 values around 0.54 and RMSE exceeding 102. While regularization in LASSO and Ridge helps reduce overfitting, their predictive improvements over basic linear regression are marginal. Although the XGBoost model's training R^2 (0.892) is slightly higher than its test R^2 (0.695), suggesting minor overfitting, it still achieves the best overall performance. This indicates that the model captures complex nonlinear relationships more effectively than other approaches.

5 How good is the prediction model?

The XGBoost model demonstrates strong predictive performance for Airbnb price prediction, as we referred last section. It suggests the model explains approximately 70% of the variance in the test set, which is substantially better than linear models such as LASSO, Ridge and Random Forest. The training R^2 is 0.892, suggesting the model fits the training data well. The gap between training and test R^2 (0.892 vs. 0.695) is reasonable, indicating some overfitting but still acceptable given the complexity of the task. Compared to Random Forest, which had similar overfitting characteristics but lower test R^2, XGBoost performs better.

The residual plot (see Appendix) displays a fairly random dispersion around zero, especially for lower predicted prices, suggesting no strong evidence of heteroscedasticity or systematic bias. While some large residuals are present at higher price ranges($\$300-\$700$), this is expected due to the lower density of listings in that range, resulting in greater variability and less model precision.

6 Further recommendations

The amenities variable was excluded due to time constraints. In future work, we could extract more value by computing the number of amenities, creating dummy variables for key features (e.g., Wi-Fi, pool), or grouping them into categories (e.g., comfort, entertainment). These engineered variables could better reflect listing quality and guest preferences.

Additionally, Airbnb pricing is sensitive to seasonal effects and special events. Incorporating calendar features like holiday flags, weekend indicators, or peak travel season could improve temporal accuracy. External data sources such as tourism statistics or Google Trends may also enhance the model by capturing broader demand patterns. If more time and resources were available, we would explore advanced techniques such as hyperparameter tuning for XGBoost using cross-validation, ensemble methods like stacking.

7 Appendix

7.1 Generative AI usage

ChatGPT was used to generate and troubleshoot R and LaTeX code, suggest ideas, and improve the grammar and clarity of my writing.

7.2 R code with useful results

A2_final

Yuan_Zhang

2025-08-04

Question 1

R preparation and data cleaning

```
#if needed{install.packages(c( "skimr","dplyr", "caret", "MASS", "geosphere", "randomForest"));install.
getwd()
```

```
## [1] "C:/Users/laptop/Desktop"
```

```
  setwd('C:/Users/laptop/Desktop/UNSW study/actl5110')
  mel_original<-read.csv("AirbnbMelbourneV2.csv",header=TRUE)
#Check the data set, convert the unusable data format
library(skimr)
library(rlang)
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.3.2
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(car)
```

```
## Warning: package 'car' was built under R version 4.3.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.3.2
```

```
##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##     recode
```

```
skim(mel_original)
```

Table 1: Data summary

Name	mel_original
Number of rows	8990
Number of columns	51
Column type frequency:	
character	18
logical	2
numeric	31
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
host_since	0	1	8	10	0	2876	0
host_response_time	0	1	12	18	0	4	0
host_response_rate	0	1	2	4	0	55	0
host_acceptance_rate	0	1	2	4	0	94	0
host_is_superhost	0	1	0	1	55	3	0
host_verifications	0	1	9	32	0	5	0
host_has_profile_pic	0	1	1	1	0	2	0
host_identity_verified	0	1	1	1	0	2	0
neighbourhood	0	1	4	17	0	30	0
property_type	0	1	3	34	0	72	0
room_type	0	1	10	15	0	4	0
bathrooms_text	0	1	0	17	1	34	0
amenities	0	1	2	2482	0	8669	0
price	0	1	6	9	0	757	0
has_availability	0	1	1	1	0	1	0
first_review	0	1	8	10	0	2431	0
last_review	0	1	8	10	0	654	0
instant_bookable	0	1	1	1	0	2	0

Variable type: logical

skim_variable	n_missing	complete_rate	mean	count
X	8990	0	NaN	:
X.1	8990	0	NaN	:

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
host_listings_count	0	1	24.03	56.69	1.00	1.00	3.00	19.00	377.00	
host_total_listings_count	0	1	38.62	101.38	1.00	2.00	5.00	29.00	794.00	
latitude	0	1	-37.83	0.08	-38.22	-37.85	-37.82	-37.80	-37.49	
longitude	0	1	145.02	0.17	144.52	144.95	144.97	145.04	145.83	
accommodates	0	1	3.93	2.47	1.00	2.00	4.00	5.00	16.00	
bathrooms	0	1	1.42	0.71	0.00	1.00	1.00	2.00	9.00	
bedrooms	0	1	1.77	1.05	0.00	1.00	2.00	2.00	10.00	
beds	0	1	2.18	1.65	0.00	1.00	2.00	3.00	18.00	
minimum_nights	0	1	3.07	8.32	1.00	1.00	2.00	3.00	365.00	
maximum_nights	0	1	488.26	422.45	1.00	90.00	365.00	1125.00	1125.00	
minimum_minimum_nights	0	1	2.71	7.43	1.00	1.00	2.00	2.00	360.00	
maximum_minimum_nights	0	1	4.77	25.40	1.00	2.00	2.00	4.00	1000.00	
minimum_maximum_nights	0	1	239527.57	22649044.63	1.00	365.00	365.00	1125.00	2147483647.00	
maximum_maximum_nights	0	1	239558.10	22649044.31	1.00	365.00	365.00	1125.00	2147483647.00	
minimum_nights_avg_ntm	0	1	3.69	10.46	1.00	1.10	2.00	3.00	360.00	
maximum_nights_avg_ntm	0	1	239542.21	22649044.48	1.00	365.00	365.00	1125.00	2147483647.00	
availability_30	0	1	13.54	9.04	0.00	6.00	14.00	22.00	30.00	
availability_60	0	1	34.17	18.30	0.00	21.00	38.00	50.00	60.00	
availability_90	0	1	56.76	27.13	0.00	40.00	65.00	79.00	90.00	
availability_365	0	1	184.49	118.87	0.00	77.00	168.00	301.00	365.00	
number_of_reviews	0	1	47.01	73.97	1.00	5.00	18.00	55.00	923.00	
number_of_reviews_ltm	0	1	15.39	18.29	0.00	3.00	9.00	22.00	286.00	
number_of_reviews_l30d	0	1	1.57	1.87	0.00	0.00	1.00	2.00	19.00	
review_scores_rating	0	1	4.71	0.44	1.00	4.65	4.85	4.97	5.00	
review_scores_accuracy	0	1	4.74	0.42	1.00	4.70	4.88	4.98	5.00	
review_scores_cleanliness	0	1	4.68	0.46	1.00	4.58	4.81	4.96	5.00	
review_scores_checkin	0	1	4.77	0.39	1.00	4.71	4.90	5.00	5.00	
review_scores_communication	0	1	4.81	0.39	1.00	4.80	4.94	5.00	5.00	
review_scores_location	0	1	4.82	0.30	1.00	4.77	4.90	5.00	5.00	
review_scores_value	0	1	4.65	0.45	1.00	4.57	4.77	4.90	5.00	
reviews_per_month	0	1	1.80	1.67	0.01	0.63	1.36	2.46	29.39	

```r
#create avg_score
names(mel_original)[41:47]
```

```
## [1] "review_scores_rating"        "review_scores_accuracy"
## [3] "review_scores_cleanliness"   "review_scores_checkin"
## [5] "review_scores_communication" "review_scores_location"
## [7] "review_scores_value"
```

```r
mel_original$avg_score <- rowMeans(mel_original[, 41:47], na.rm = TRUE)
#create binary, 'HAS=1'or'HAS=0'
mel_original$HAS <- ifelse(mel_original$avg_score > 4.9, 1, 0)

#data cleaning
# Remove review-related columns due to we cannot use 'review_score_' as the predictors
mel_original <- mel_original[, !grepl("review_scores", names(mel_original))]
mel<- subset(mel_original, select = -c(avg_score))
```

```r
# Convert character percentages to numeric
mel$host_response_rate <- as.numeric(gsub("%", "", mel$host_response_rate))
mel$host_acceptance_rate <- as.numeric(gsub("%", "", mel$host_acceptance_rate))

# Clean price
mel$price <- as.numeric(gsub("[$,]", "", mel$price))

# Convert logical categorical variables into factor!
mel$host_is_superhost <- as.factor(mel$host_is_superhost)
mel$instant_bookable <- as.factor(mel$instant_bookable)
mel$host_has_profile_pic <- as.factor(mel$host_has_profile_pic)
mel$host_identity_verified <- as.factor(mel$host_identity_verified)

# Convert character variables with few levels into factors
char_vars <- sapply(mel, is.character)
mel[char_vars] <- lapply(mel[char_vars], as.factor)

# convert 'latitude' and 'longtitude' into a new variable
melbourne_cbd_lat <- -37.8124
melbourne_cbd_lon <- 144.9623
library(geosphere)
```

```
## Warning: package 'geosphere' was built under R version 4.3.3
```

```r
## Compute distance from each listing to Melbourne CBD in meters
coords_listings <- cbind(mel$longitude, mel$latitude)
coords_cbd <- c(melbourne_cbd_lon, melbourne_cbd_lat)
## Distance in meters
mel$distance_meters <- distHaversine(coords_listings, coords_cbd)
## Convert to kilometers
mel$distance_km <- mel$distance_meters / 1000
## Categorize distance
mel$distance_group <- cut(
  mel$distance_km,
  breaks = c(-Inf, 2, 5, 10, Inf),
  labels = c("0-2 km", "2-5 km", "5-10 km", ">10 km"),
  right = FALSE
)

# convert host_since to numerical variable
## Set reference date
today <- as.Date("2025-07-29")

# Convert host_since from factor to Date
mel$host_since <- as.Date(as.character(mel$host_since), format = "%Y/%m/%d")

# Create new numeric variable: years since hosting
mel$host_since_years <- as.numeric(difftime(today, mel$host_since, units = "days")) / 365

# check accommodates > 0
mel <- mel %>% filter(accommodates > 0)

# bedroom and bathroom_per_person
```

```r
mel$bathroom_per_person <- mel$bathrooms / mel$accommodates
mel$bedroom_per_person  <- mel$bedrooms / mel$accommodates

# Convert to character
mel$host_is_superhost <- as.character(mel$host_is_superhost)

# Replace "" with "f"
mel$host_is_superhost[mel$host_is_superhost == ""] <- "f"

# Convert back to factor
mel$host_is_superhost <- as.factor(mel$host_is_superhost)

#add on interaction and transformation

str(mel) # now, all character variables are converted into factor
```

```
## 'data.frame':    8990 obs. of  51 variables:
##  $ host_since                : Date, format: "2022-02-24" "2020-12-30" ...
##  $ host_response_time        : Factor w/ 4 levels "a few days or more",..: 4 3 4 4 4 3 4 4 4 3 ...
##  $ host_response_rate        : num  95 89 100 100 97 82 100 100 100 100 ...
##  $ host_acceptance_rate      : num  99 57 99 100 94 41 100 100 100 56 ...
##  $ host_is_superhost         : Factor w/ 2 levels "f","t": 1 1 1 1 1 1 1 1 2 1 ...
##  $ host_listings_count       : int  8 3 109 1 7 346 2 1 4 1 ...
##  $ host_total_listings_count: int  8 4 270 2 7 368 3 1 6 3 ...
##  $ host_verifications        : Factor w/ 5 levels "['email', 'phone', 'work_email']",..: 2 5 1 1 2 1
##  $ host_has_profile_pic      : Factor w/ 2 levels "f","t": 2 2 2 2 2 2 2 2 2 2 ...
##  $ host_identity_verified    : Factor w/ 2 levels "f","t": 2 2 2 2 2 2 2 2 2 2 ...
##  $ neighbourhood             : Factor w/ 30 levels "Banyule","Bayside",..: 24 4 18 9 24 18 29 18 29 9
##  $ latitude                  : num  -37.9 -37.8 -37.8 -37.9 -37.8 ...
##  $ longitude                 : num  145 145 145 145 145 ...
##  $ property_type             : Factor w/ 72 levels "Barn","Boat",..: 63 39 19 39 19 39 39 21 19 44 ..
##  $ room_type                 : Factor w/ 4 levels "Entire home/apt",..: 4 3 1 3 1 3 3 1 1 3 ...
##  $ accommodates              : int  1 1 3 1 2 2 2 3 4 2 ...
##  $ bathrooms                 : num  1 1 1 1 1 2 1 1 1 1 ...
##  $ bathrooms_text            : Factor w/ 34 levels "","0 baths","0 shared baths",..: 6 6 4 4 4 10 5 4
##  $ bedrooms                  : int  1 1 2 1 1 1 1 1 1 1 ...
##  $ beds                      : int  1 0 2 1 1 2 2 2 2 0 ...
##  $ amenities                 : Factor w/ 8669 levels "[\"55\\\" HDTV with Amazon Prime Video, Disney+
##  $ price                     : num  123 50 166 120 202 80 75 104 175 55 ...
##  $ minimum_nights            : int  1 1 3 1 2 7 1 5 2 2 ...
##  $ maximum_nights            : int  365 365 90 365 365 365 1125 365 365 1125 ...
##  $ minimum_minimum_nights    : int  1 1 3 1 2 7 1 2 3 2 ...
##  $ maximum_minimum_nights    : int  1 1 3 1 2 7 1 5 3 2 ...
##  $ minimum_maximum_nights    : int  30 365 1125 365 365 365 1125 365 365 1125 ...
##  $ maximum_maximum_nights    : int  1125 365 1125 365 365 365 1125 365 365 1125 ...
##  $ minimum_nights_avg_ntm    : num  1 1 3 1 2 7 1 5 3 2 ...
##  $ maximum_nights_avg_ntm    : num  262 365 1125 365 365 ...
##  $ has_availability          : Factor w/ 1 level "t": 1 1 1 1 1 1 1 1 1 1 ...
##  $ availability_30           : int  29 24 12 29 0 4 6 0 13 20 ...
##  $ availability_60           : int  59 54 42 59 0 34 24 0 13 50 ...
##  $ availability_90           : int  89 84 72 89 0 64 53 20 13 80 ...
##  $ availability_365          : int  352 359 88 269 0 64 233 215 13 355 ...
##  $ number_of_reviews         : int  30 5 2 1 2 2 100 5 19 11 ...
```

```
##  $ number_of_reviews_ltm    : int  9 5 2 1 2 2 15 5 19 6 ...
##  $ number_of_reviews_l30d   : int  1 1 1 1 2 0 6 3 0 1 ...
##  $ first_review             : Factor w/ 2431 levels "2010/11/24","2010/12/5",..: 1814 2332 2365 2429
##  $ last_review              : Factor w/ 654 levels "2014/3/10","2016/12/3",..: 616 616 626 652 637 5
##  $ instant_bookable         : Factor w/ 2 levels "f","t": 2 1 1 1 1 1 2 2 2 1 ...
##  $ reviews_per_month        : num  1.21 0.81 0.98 1 2 0.17 1.08 3.85 1.75 0.68 ...
##  $ X                        : logi  NA NA NA NA NA NA ...
##  $ X.1                      : logi  NA NA NA NA NA NA ...
##  $ HAS                      : num  0 0 0 1 0 0 0 1 0 1 ...
##  $ distance_meters          : num  5444 10721 1186 15217 2404 ...
##  $ distance_km              : num  5.44 10.72 1.19 15.22 2.4 ...
##  $ distance_group           : Factor w/ 4 levels "0-2 km","2-5 km",..: 3 4 1 4 2 1 2 2 1 4 ...
##  $ host_since_years         : num  3.43 4.58 10.78 11.16 1.61 ...
##  $ bathroom_per_person      : num  1 1 0.333 1 0.5 ...
##  $ bedroom_per_person       : num  1 1 0.667 1 0.5 ...
```

Split data into train and test

```r
library(dplyr)

set.seed(123)

#choose 70% of data as training data
train_1 <- mel %>% filter(HAS == 1) %>% slice_sample(prop = 0.7)
train_0 <- mel %>% filter(HAS == 0) %>% slice_sample(prop = 0.7)

# combine HAS=1 AND 0
train <- bind_rows(train_1, train_0)

# the remaining are the test data set
test <- anti_join(mel, train)
```

```
## Joining with `by = join_by(host_since, host_response_time, host_response_rate,
## host_acceptance_rate, host_is_superhost, host_listings_count,
## host_total_listings_count, host_verifications, host_has_profile_pic,
## host_identity_verified, neighbourhood, latitude, longitude, property_type,
## room_type, accommodates, bathrooms, bathrooms_text, bedrooms, beds, amenities,
## price, minimum_nights, maximum_nights, minimum_minimum_nights,
## maximum_minimum_nights, minimum_maximum_nights, maximum_maximum_nights,
## minimum_nights_avg_ntm, maximum_nights_avg_ntm, has_availability,
## availability_30, availability_60, availability_90, availability_365,
## number_of_reviews, number_of_reviews_ltm, number_of_reviews_l30d, first_review,
## last_review, instant_bookable, reviews_per_month, X, X.1, HAS, distance_meters,
## distance_km, distance_group, host_since_years, bathroom_per_person,
## bedroom_per_person)`
```

```r
# check results
table(train$HAS)
```

```
##
##    0    1
## 4182 2110
```

```r
table(test$HAS)
```

```
##
##    0    1
## 1793  905
```

```r
# check proportions
prop.table(table(mel$HAS))
```

```
##
##         0         1
## 0.6646274 0.3353726
```

```r
prop.table(table(train$HAS))
```

```
##
##         0         1
## 0.6646535 0.3353465
```

```r
prop.table(table(test$HAS)) #the proportions are close which are acceptable
```

```
##
##         0         1
## 0.6645663 0.3354337
```

Fit models

Model1: LASSO

```r
#LASSO
# Start with full model, delete the columns which are with too many factors and the dates
drop_cols <- c("amenities", "bathrooms_text",
               "has_availability", "host_since","property_type","first_review","last_review","X","X.1")
mel_lasso <- mel[, !(names(mel) %in% drop_cols)]
str(mel_lasso)
```

```
## 'data.frame':    8990 obs. of  42 variables:
##  $ host_response_time       : Factor w/ 4 levels "a few days or more",..: 4 3 4 4 4 3 4 4 4 3 ...
##  $ host_response_rate       : num  95 89 100 100 97 82 100 100 100 100 ...
##  $ host_acceptance_rate     : num  99 57 99 100 94 41 100 100 100 56 ...
##  $ host_is_superhost        : Factor w/ 2 levels "f","t": 1 1 1 1 1 1 1 1 2 1 ...
##  $ host_listings_count      : int  8 3 109 1 7 346 2 1 4 1 ...
##  $ host_total_listings_count: int  8 4 270 2 7 368 3 1 6 3 ...
##  $ host_verifications       : Factor w/ 5 levels "['email', 'phone', 'work_email']",..: 2 5 1 1 2 1
##  $ host_has_profile_pic     : Factor w/ 2 levels "f","t": 2 2 2 2 2 2 2 2 2 2 ...
##  $ host_identity_verified   : Factor w/ 2 levels "f","t": 2 2 2 2 2 2 2 2 2 2 ...
##  $ neighbourhood            : Factor w/ 30 levels "Banyule","Bayside",..: 24 4 18 9 24 18 29 18 29 9
##  $ latitude                 : num  -37.9 -37.8 -37.8 -37.9 -37.8 ...
```

```
##  $ longitude                 : num  145 145 145 145 145 ...
##  $ room_type                 : Factor w/ 4 levels "Entire home/apt",..: 4 3 1 3 1 3 3 1 1 3 ...
##  $ accommodates              : int  1 1 3 1 2 2 2 3 4 2 ...
##  $ bathrooms                 : num  1 1 1 1 1 2 1 1 1 1 ...
##  $ bedrooms                  : int  1 1 2 1 1 1 1 1 1 1 ...
##  $ beds                      : int  1 0 2 1 1 2 2 2 2 0 ...
##  $ price                     : num  123 50 166 120 202 80 75 104 175 55 ...
##  $ minimum_nights            : int  1 1 3 1 2 7 1 5 2 2 ...
##  $ maximum_nights            : int  365 365 90 365 365 365 1125 365 365 1125 ...
##  $ minimum_minimum_nights    : int  1 1 3 1 2 7 1 2 3 2 ...
##  $ maximum_minimum_nights    : int  1 1 3 1 2 7 1 5 3 2 ...
##  $ minimum_maximum_nights    : int  30 365 1125 365 365 365 1125 365 365 1125 ...
##  $ maximum_maximum_nights    : int  1125 365 1125 365 365 365 1125 365 365 1125 ...
##  $ minimum_nights_avg_ntm    : num  1 1 3 1 2 7 1 5 3 2 ...
##  $ maximum_nights_avg_ntm    : num  262 365 1125 365 365 ...
##  $ availability_30           : int  29 24 12 29 0 4 6 0 13 20 ...
##  $ availability_60           : int  59 54 42 59 0 34 24 0 13 50 ...
##  $ availability_90           : int  89 84 72 89 0 64 53 20 13 80 ...
##  $ availability_365          : int  352 359 88 269 0 64 233 215 13 355 ...
##  $ number_of_reviews         : int  30 5 2 1 2 2 100 5 19 11 ...
##  $ number_of_reviews_ltm     : int  9 5 2 1 2 2 15 5 19 6 ...
##  $ number_of_reviews_l30d    : int  1 1 1 1 2 0 6 3 0 1 ...
##  $ instant_bookable          : Factor w/ 2 levels "f","t": 2 1 1 1 1 1 2 2 2 1 ...
##  $ reviews_per_month         : num  1.21 0.81 0.98 1 2 0.17 1.08 3.85 1.75 0.68 ...
##  $ HAS                       : num  0 0 0 1 0 0 0 1 0 1 ...
##  $ distance_meters           : num  5444 10721 1186 15217 2404 ...
##  $ distance_km               : num  5.44 10.72 1.19 15.22 2.4 ...
##  $ distance_group            : Factor w/ 4 levels "0-2 km","2-5 km",..: 3 4 1 4 2 1 2 2 1 4 ...
##  $ host_since_years          : num  3.43 4.58 10.78 11.16 1.61 ...
##  $ bathroom_per_person       : num  1 1 0.333 1 0.5 ...
##  $ bedroom_per_person        : num  1 1 0.667 1 0.5 ...
```

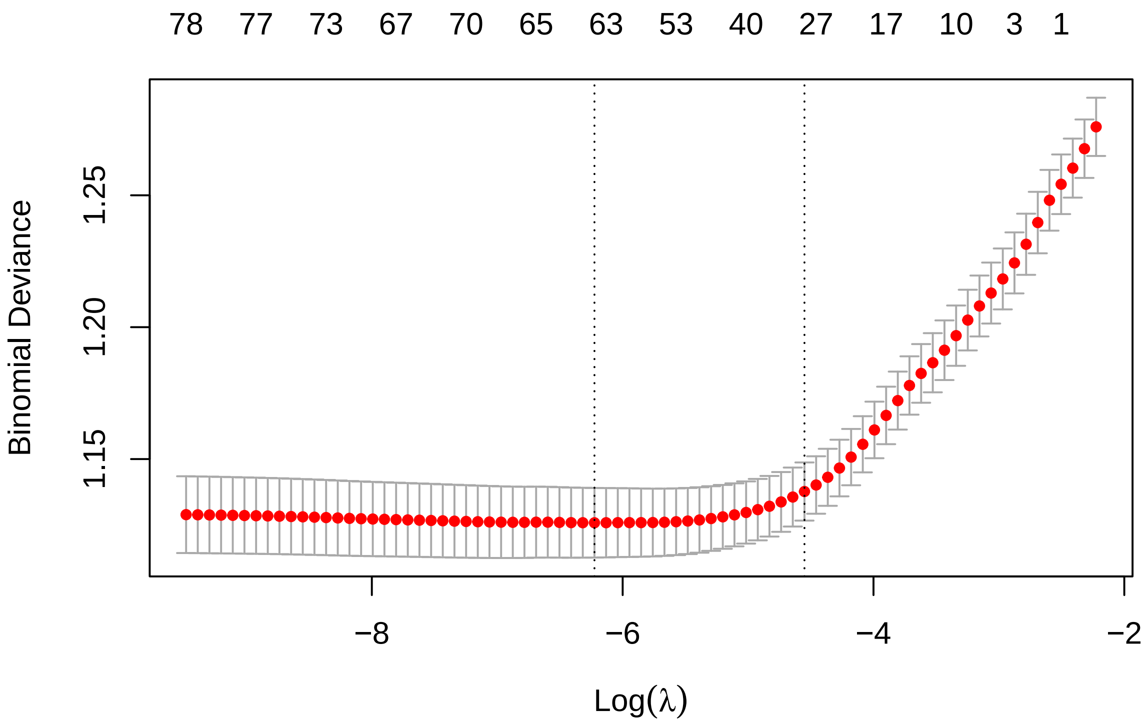
```
#categorical with encoding one-hot
X <- model.matrix(HAS ~ . -1, data = mel_lasso)  # -1 delete Intercept
y <- mel_lasso$HAS

train_clean <- train[, !(names(train) %in% drop_cols)]
test_clean <- test[, !(names(test) %in% drop_cols)]

# Convert factor variables to dummy variables (one-hot encoding)
x_train <- model.matrix(HAS ~ ., data = train_clean)[, -1]  # remove intercept
y_train <- as.factor(train_clean$HAS)

x_test <- model.matrix(HAS ~ ., data = test_clean)[, -1]
y_test <- as.factor(test_clean$HAS)

library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.3.3
```

```
## Loading required package: Matrix
```

```
## Loaded glmnet 4.1-8
```

```r
#fit lasso model
set.seed(123)
lasso_cv <- cv.glmnet(x_train, y_train, family = "binomial", alpha = 1)
best_lambda <- lasso_cv$lambda.min


## Plot cross-validation
plot(lasso_cv)
```




```r
##Lambda with the lowest cross-validated error
best_lambda
```

```
## [1] 0.001979324
```

```r
#get coefficient from lasso
lasso_coef <- coef(lasso_cv, s = "lambda.min")
print(lasso_coef)
```

```
## 79 x 1 sparse Matrix of class "dgCMatrix"
##                                                        s1
## (Intercept)                                  -3.055450e-01
## host_response_timewithin a day                .
## host_response_timewithin a few hours          6.184702e-02
## host_response_timewithin an hour             -3.478063e-02
```

```
## host_response_rate                          5.123059e-03
## host_acceptance_rate                       -8.774922e-03
## host_is_superhostt                          9.569001e-01
## host_listings_count                        -2.125822e-03
## host_total_listings_count                  -3.176991e-03
## host_verifications['email', 'phone']        3.194189e-01
## host_verifications['email']                -1.074149e+00
## host_verifications['phone', 'work_email'] -4.323544e-01
## host_verifications['phone']                 1.313014e-01
## host_has_profile_pict                      -3.048388e-01
## host_identity_verifiedt                    -3.512084e-01
## neighbourhoodBayside                        1.407521e-01
## neighbourhoodBoroondara                     .
## neighbourhoodBrimbank                       .
## neighbourhoodCardinia                       1.080757e+00
## neighbourhoodCasey                          1.348880e-01
## neighbourhoodDarebin                        .
## neighbourhoodFrankston                      7.136805e-02
## neighbourhoodGlen Eira                      3.336266e-01
## neighbourhoodGreater Dandenong             -2.035455e-01
## neighbourhoodHobsons Bay                   -2.670910e-02
## neighbourhoodHume                           5.062411e-01
## neighbourhoodKingston                       8.048652e-02
## neighbourhoodKnox                          -1.132196e-01
## neighbourhoodManningham                    -2.362983e-01
## neighbourhoodMaribyrnong                   -4.484805e-01
## neighbourhoodMaroondah                      4.847579e-02
## neighbourhoodMelbourne                     -4.399729e-01
## neighbourhoodMelton                         2.958468e-02
## neighbourhoodMonash                        -2.285821e-01
## neighbourhoodMoonee Valley                  1.159614e-01
## neighbourhoodMoreland                       1.282100e-01
## neighbourhoodNillumbik                      2.826590e-01
## neighbourhoodPort Phillip                  -4.909469e-02
## neighbourhoodStonnington                    1.401858e-02
## neighbourhoodWhitehorse                    -6.390813e-02
## neighbourhoodWhittlesea                    -1.038873e-01
## neighbourhoodWyndham                        .
## neighbourhoodYarra                          2.322731e-02
## neighbourhoodYarra Ranges                   2.846380e-01
## latitude                                    .
## longitude                                   .
## room_typeHotel room                        -1.702492e+00
## room_typePrivate room                       4.726673e-01
## room_typeShared room                       -2.317402e-01
## accommodates                                .
## bathrooms                                   .
## bedrooms                                    .
## beds                                       -1.158285e-01
## price                                       1.938390e-03
## minimum_nights                              4.463242e-03
## maximum_nights                             -3.221142e-04
## minimum_minimum_nights                     -4.863408e-03
## maximum_minimum_nights                      .
```

```
## minimum_maximum_nights                                  .
## maximum_maximum_nights                      -6.455678e-05
## minimum_nights_avg_ntm                      -6.002095e-03
## maximum_nights_avg_ntm                                  .
## availability_30                             -1.122720e-02
## availability_60                                         .
## availability_90                             -5.792841e-04
## availability_365                                        .
## number_of_reviews                           -4.257262e-04
## number_of_reviews_ltm                       -1.565411e-02
## number_of_reviews_l30d                       1.208511e-01
## instant_bookablet                           -2.460759e-01
## reviews_per_month                            5.542192e-02
## distance_meters                              6.335617e-06
## distance_km                                  1.135709e-16
## distance_group2-5 km                         1.093604e-01
## distance_group5-10 km                                   .
## distance_group>10 km                        -7.643270e-02
## host_since_years                             1.540738e-02
## bathroom_per_person                         -7.107116e-02
## bedroom_per_person                           1.338060e-01
```

```
lasso_cv$lambda.min
```

```
## [1] 0.001979324
```

```
# Extract the names of variables selected by LASSO
selected_vars <- rownames(lasso_coef)[which(lasso_coef != 0)]
selected_vars <- selected_vars[selected_vars != "(Intercept)"]  # exclude intercept

x_train_selected <- x_train[, selected_vars]
logit_model <- glm(y_train ~ ., data = as.data.frame(x_train_selected), family = "binomial")
x_test_selected<-x_test[,selected_vars]

# View summary with p-values and coefficients
summary(logit_model)
```

```
##
## Call:
## glm(formula = y_train ~ ., family = "binomial", data = as.data.frame(x_train_selected))
##
## Coefficients: (1 not defined because of singularities)
##                                             Estimate Std. Error z value
## (Intercept)                               -2.181e-01  4.056e-01  -0.538
## `host_response_timewithin a few hours`     5.915e-02  1.561e-01   0.379
## `host_response_timewithin an hour`        -4.874e-02  1.492e-01  -0.327
## host_response_rate                         6.758e-03  3.050e-03   2.216
## host_acceptance_rate                      -1.013e-02  2.203e-03  -4.600
## host_is_superhostt                         1.005e+00  6.788e-02  14.805
## host_listings_count                       -1.382e-03  2.175e-03  -0.635
## host_total_listings_count                 -3.822e-03  1.506e-03  -2.537
## `host_verifications['email', 'phone']`     3.536e-01  9.022e-02   3.919
## `host_verifications['email']`             -1.178e+01  2.110e+02  -0.056
```

```
## `host_verifications['phone', 'work_email']` -5.619e-01  4.107e-01  -1.368
## `host_verifications['phone']`               2.152e-01  1.342e-01   1.604
## host_has_profile_pict                       -3.905e-01  2.188e-01  -1.785
## host_identity_verifiedt                     -3.740e-01  1.628e-01  -2.297
## neighbourhoodBayside                         2.948e-01  2.570e-01   1.147
## neighbourhoodCardinia                        1.359e+00  4.094e-01   3.320
## neighbourhoodCasey                           3.503e-01  3.082e-01   1.137
## neighbourhoodFrankston                       2.803e-01  2.909e-01   0.964
## `neighbourhoodGlen Eira`                     4.202e-01  2.031e-01   2.069
## `neighbourhoodGreater Dandenong`            -2.113e-01  3.691e-01  -0.572
## `neighbourhoodHobsons Bay`                  -1.327e-01  2.699e-01  -0.491
## neighbourhoodHume                            7.013e-01  2.772e-01   2.530
## neighbourhoodKingston                        2.850e-01  2.529e-01   1.127
## neighbourhoodKnox                           -9.656e-02  3.744e-01  -0.258
## neighbourhoodManningham                     -2.232e-01  3.510e-01  -0.636
## neighbourhoodMaribyrnong                    -6.120e-01  2.337e-01  -2.619
## neighbourhoodMaroondah                       3.750e-01  4.222e-01   0.888
## neighbourhoodMelbourne                      -5.477e-01  1.467e-01  -3.734
## neighbourhoodMelton                          2.625e-01  3.320e-01   0.791
## neighbourhoodMonash                         -1.852e-01  2.389e-01  -0.775
## `neighbourhoodMoonee Valley`                 1.331e-01  2.506e-01   0.531
## neighbourhoodMoreland                        1.157e-01  1.936e-01   0.598
## neighbourhoodNillumbik                       5.420e-01  3.812e-01   1.422
## `neighbourhoodPort Phillip`                 -1.730e-01  1.550e-01  -1.116
## neighbourhoodStonnington                    -2.261e-02  1.768e-01  -0.128
## neighbourhoodWhitehorse                      1.882e-03  2.222e-01   0.008
## neighbourhoodWhittlesea                     -1.107e-01  3.389e-01  -0.327
## neighbourhoodYarra                          -4.109e-02  1.812e-01  -0.227
## `neighbourhoodYarra Ranges`                  4.710e-01  2.686e-01   1.754
## `room_typeHotel room`                       -1.249e+01  1.938e+02  -0.064
## `room_typePrivate room`                      5.469e-01  9.178e-02   5.958
## `room_typeShared room`                      -2.930e-01  3.816e-01  -0.768
## beds                                        -1.290e-01  2.394e-02  -5.389
## price                                        2.134e-03  2.382e-04   8.957
## minimum_nights                               3.005e-02  1.458e-02   2.061
## maximum_nights                              -3.332e-04  8.358e-05  -3.986
## minimum_minimum_nights                      -2.668e-02  1.490e-02  -1.790
## maximum_maximum_nights                      -6.470e-05  7.376e-05  -0.877
## minimum_nights_avg_ntm                      -1.327e-02  1.078e-02  -1.231
## availability_30                             -1.261e-02  5.483e-03  -2.299
## availability_90                             -5.187e-04  1.763e-03  -0.294
## number_of_reviews                           -4.737e-04  5.524e-04  -0.858
## number_of_reviews_ltm                       -1.901e-02  3.220e-03  -5.905
## number_of_reviews_l30d                       1.276e-01  2.363e-02   5.400
## instant_bookablet                           -2.636e-01  7.388e-02  -3.567
## reviews_per_month                            8.500e-02  3.304e-02   2.573
## distance_meters                              5.116e-06  7.076e-06   0.723
## distance_km                                         NA         NA      NA
## `distance_group2-5 km`                       1.131e-01  9.272e-02   1.220
## `distance_group>10 km`                      -2.588e-01  1.504e-01  -1.721
## host_since_years                             1.908e-02  1.038e-02   1.839
## bathroom_per_person                         -1.690e-01  1.161e-01  -1.456
## bedroom_per_person                           1.977e-01  1.401e-01   1.412
##                                             Pr(>|z|)
```

```
## (Intercept)                                 0.590860
## `host_response_timewithin a few hours`      0.704746
## `host_response_timewithin an hour`          0.743921
## host_response_rate                          0.026707 *
## host_acceptance_rate                        4.23e-06 ***
## host_is_superhostt                           < 2e-16 ***
## host_listings_count                         0.525170
## host_total_listings_count                   0.011188 *
## `host_verifications['email', 'phone']`      8.88e-05 ***
## `host_verifications['email']`               0.955461
## `host_verifications['phone', 'work_email']` 0.171237
## `host_verifications['phone']`               0.108733
## host_has_profile_pict                       0.074291 .
## host_identity_verifiedt                     0.021626 *
## neighbourhoodBayside                        0.251400
## neighbourhoodCardinia                       0.000901 ***
## neighbourhoodCasey                          0.255684
## neighbourhoodFrankston                      0.335273
## `neighbourhoodGlen Eira`                    0.038532 *
## `neighbourhoodGreater Dandenong`            0.566996
## `neighbourhoodHobsons Bay`                  0.623075
## neighbourhoodHume                           0.011411 *
## neighbourhoodKingston                       0.259789
## neighbourhoodKnox                           0.796471
## neighbourhoodManningham                     0.524930
## neighbourhoodMaribyrnong                    0.008815 **
## neighbourhoodMaroondah                      0.374475
## neighbourhoodMelbourne                      0.000189 ***
## neighbourhoodMelton                         0.429164
## neighbourhoodMonash                         0.438151
## `neighbourhoodMoonee Valley`                0.595290
## neighbourhoodMoreland                       0.549951
## neighbourhoodNillumbik                      0.155081
## `neighbourhoodPort Phillip`                 0.264396
## neighbourhoodStonnington                    0.898205
## neighbourhoodWhitehorse                     0.993242
## neighbourhoodWhittlesea                     0.743910
## neighbourhoodYarra                          0.820589
## `neighbourhoodYarra Ranges`                 0.079466 .
## `room_typeHotel room`                       0.948620
## `room_typePrivate room`                     2.55e-09 ***
## `room_typeShared room`                      0.442525
## beds                                        7.08e-08 ***
## price                                        < 2e-16 ***
## minimum_nights                              0.039274 *
## maximum_nights                              6.72e-05 ***
## minimum_minimum_nights                      0.073407 .
## maximum_maximum_nights                      0.380427
## minimum_nights_avg_ntm                      0.218243
## availability_30                             0.021496 *
## availability_90                             0.768574
## number_of_reviews                           0.391168
## number_of_reviews_ltm                       3.53e-09 ***
## number_of_reviews_l30d                      6.66e-08 ***
```

```
## instant_bookablet                              0.000360 ***
## reviews_per_month                              0.010080 *
## distance_meters                                0.469667
## distance_km                                          NA
## `distance_group2-5 km`                         0.222599
## `distance_group>10 km`                         0.085335 .
## host_since_years                               0.065931 .
## bathroom_per_person                            0.145410
## bedroom_per_person                             0.158078
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 8027.3  on 6291  degrees of freedom
## Residual deviance: 6939.4  on 6230  degrees of freedom
## AIC: 7063.4
##
## Number of Fisher Scoring iterations: 12
```

```r
#confusion matrix
## Predict on test

pred_prob <- predict(logit_model, newdata = as.data.frame(x_test_selected), type = "response")
```

```
## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type == :
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful cases
```

```r
pred_class <- ifelse(pred_prob > 0.5, 1, 0)
conf_mat<-table(Predicted = pred_class, Actual = test$HAS)
library(pROC)
```

```
## Warning: package 'pROC' was built under R version 4.3.3
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
## Attaching package: 'pROC'
```

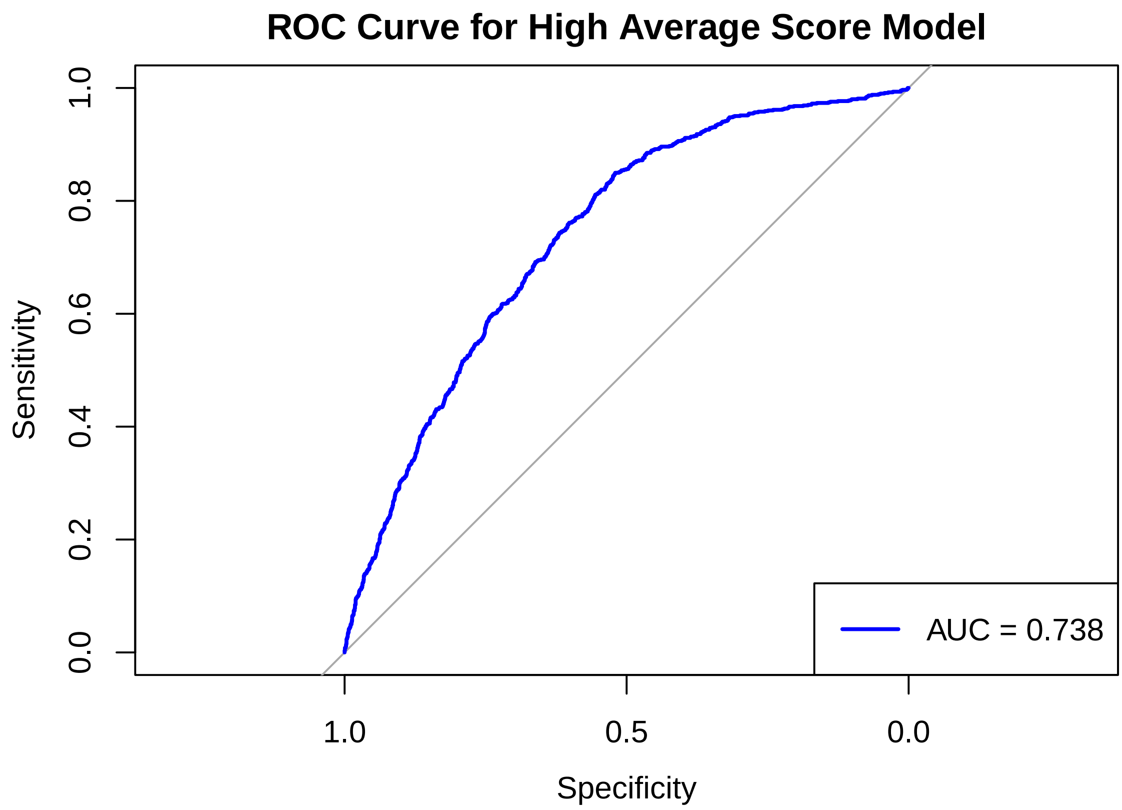
```
## The following objects are masked from 'package:stats':
##
##     cov, smooth, var
```

```r
roc_obj <- roc(test$HAS, pred_prob)
```

```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```r
plot(roc_obj, col = "blue", lwd = 2, main = "ROC Curve for High Average Score Model")
# AUC
auc_value <- auc(roc_obj)
legend("bottomright", legend = paste("AUC =", round(auc_value, 3)), col = "blue", lwd = 2)
```




```r
##acuracy
accuracy <- sum(diag(conf_mat)) / sum(conf_mat)
print(paste("Accuracy:", round(accuracy, 4)))
```

```
## [1] "Accuracy: 0.7024"
```

Model2: Significant variables from LASSO

```r
result1<-glm(train_clean$HAS ~ host_acceptance_rate +
host_is_superhost +
host_verifications +
neighbourhood+
room_type +
beds +
price +
maximum_nights +
number_of_reviews_ltm +
number_of_reviews_l30d +
```

```r
instant_bookable, data=train_clean, family = "binomial")

p1 <- predict(result1, newdata = test_clean, type = "response")
pc <- ifelse(p1 > 0.5, 1, 0)
confusion_matrix_1 <- table(Predicted = pc, Actual = test_clean$HAS)
print(confusion_matrix_1)
```

```
##          Actual
## Predicted    0    1
##         0 1562  617
##         1  231  288
```

```r
accuracy_1 <- mean(pc == test_clean$HAS)
print(paste("Accuracy:", round(accuracy_1, 4)))
```

```
## [1] "Accuracy: 0.6857"
```

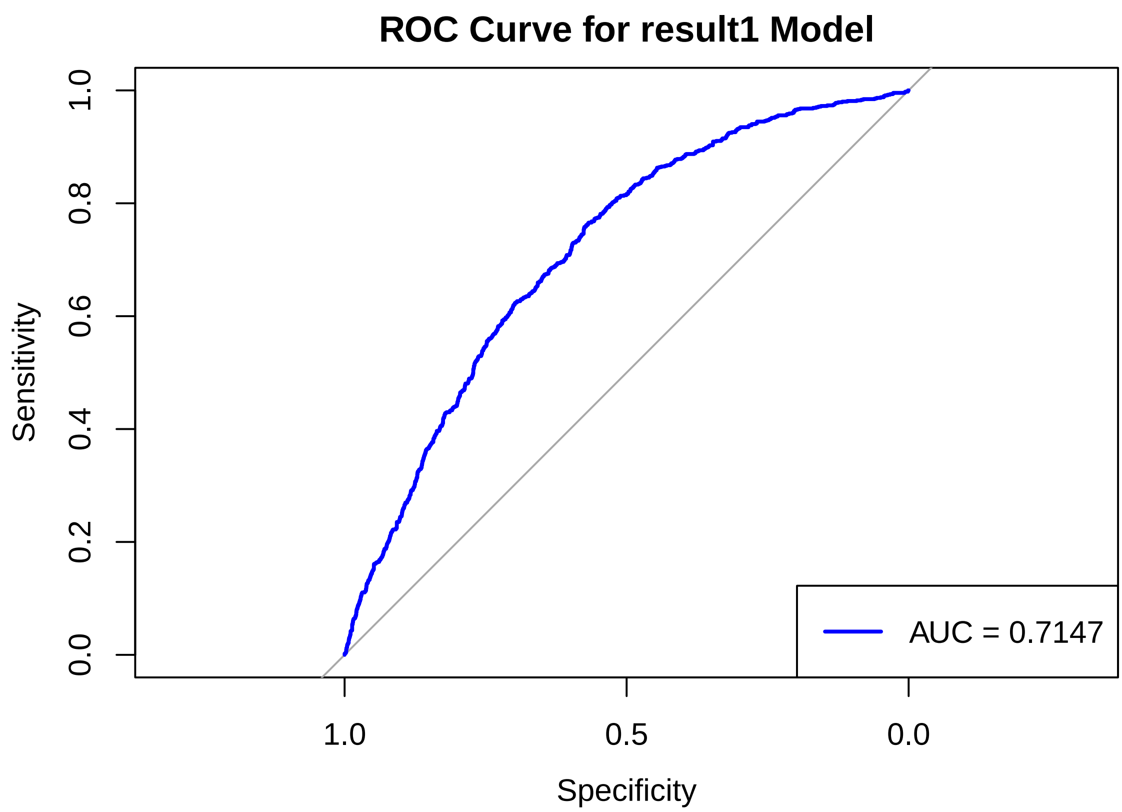
```r
# ROC
roc <- roc(response = test_clean$HAS, predictor = p1)
```

```
## Setting levels: control = 0, case = 1

## Setting direction: controls < cases
```

```r
plot(roc, col = "blue", lwd = 2, main = "ROC Curve for result1 Model")

# AUC
auc <- auc(roc)
print(paste("AUC:", round(auc, 4)))
```

```
## [1] "AUC: 0.7147"
```

```r
legend("bottomright", legend = paste("AUC =", round(auc, 4)), col = "blue", lwd = 2)
```




```r
summary(result1)
```

```
##
## Call:
## glm(formula = train_clean$HAS ~ host_acceptance_rate + host_is_superhost +
##     host_verifications + neighbourhood + room_type + beds + price +
##     maximum_nights + number_of_reviews_ltm + number_of_reviews_l30d +
##     instant_bookable, family = "binomial", data = train_clean)
##
## Coefficients:
##                                            Estimate Std. Error z value
## (Intercept)                              -1.141e+00  3.080e-01  -3.706
## host_acceptance_rate                     -6.871e-03  1.700e-03  -4.041
## host_is_superhostt                        1.080e+00  6.542e-02  16.516
## host_verifications['email', 'phone']      6.081e-01  8.265e-02   7.358
## host_verifications['email']              -1.180e+01  2.114e+02  -0.056
## host_verifications['phone', 'work_email'] -3.907e-01  4.047e-01  -0.965
## host_verifications['phone']               3.638e-01  1.233e-01   2.952
## neighbourhoodBayside                      4.269e-01  3.461e-01   1.234
## neighbourhoodBoroondara                   1.687e-01  3.104e-01   0.544
## neighbourhoodBrimbank                     1.142e-01  3.545e-01   0.322
## neighbourhoodCardinia                     1.529e+00  4.032e-01   3.792
## neighbourhoodCasey                        5.496e-01  3.466e-01   1.586
## neighbourhoodDarebin                      1.794e-01  3.245e-01   0.553
## neighbourhoodFrankston                    4.667e-01  3.386e-01   1.378
```

```
## neighbourhoodGlen Eira                      5.223e-01  3.129e-01   1.669
## neighbourhoodGreater Dandenong             -1.631e-01  4.365e-01  -0.374
## neighbourhoodHobsons Bay                    7.614e-02  3.593e-01   0.212
## neighbourhoodHume                           7.904e-01  3.606e-01   2.192
## neighbourhoodKingston                       3.519e-01  3.391e-01   1.038
## neighbourhoodKnox                          -4.671e-02  4.328e-01  -0.108
## neighbourhoodManningham                    -2.158e-01  4.216e-01  -0.512
## neighbourhoodMaribyrnong                   -3.482e-01  3.247e-01  -1.072
## neighbourhoodMaroondah                      3.384e-01  4.742e-01   0.714
## neighbourhoodMelbourne                     -4.498e-01  2.637e-01  -1.706
## neighbourhoodMelton                         3.466e-01  3.968e-01   0.874
## neighbourhoodMonash                        -1.297e-01  3.340e-01  -0.388
## neighbourhoodMoonee Valley                  3.639e-01  3.395e-01   1.072
## neighbourhoodMoreland                       2.250e-01  2.946e-01   0.764
## neighbourhoodNillumbik                      6.139e-01  4.374e-01   1.404
## neighbourhoodPort Phillip                   9.198e-02  2.714e-01   0.339
## neighbourhoodStonnington                    2.491e-01  2.848e-01   0.875
## neighbourhoodWhitehorse                     1.269e-02  3.223e-01   0.039
## neighbourhoodWhittlesea                     6.694e-02  4.114e-01   0.163
## neighbourhoodWyndham                        1.786e-01  3.061e-01   0.584
## neighbourhoodYarra                          3.269e-01  2.792e-01   1.171
## neighbourhoodYarra Ranges                   6.203e-01  2.803e-01   2.213
## room_typeHotel room                        -1.243e+01  1.934e+02  -0.064
## room_typePrivate room                       4.787e-01  8.332e-02   5.746
## room_typeShared room                       -4.512e-01  3.781e-01  -1.193
## beds                                       -1.405e-01  2.324e-02  -6.047
## price                                       2.090e-03  2.311e-04   9.043
## maximum_nights                             -3.908e-04  6.980e-05  -5.599
## number_of_reviews_ltm                      -1.625e-02  2.369e-03  -6.860
## number_of_reviews_l30d                      1.733e-01  2.092e-02   8.284
## instant_bookablet                          -3.354e-01  7.178e-02  -4.672
##                                            Pr(>|z|)
## (Intercept)                                0.000211 ***
## host_acceptance_rate                       5.33e-05 ***
## host_is_superhostt                          < 2e-16 ***
## host_verifications['email', 'phone']       1.86e-13 ***
## host_verifications['email']                0.955496
## host_verifications['phone', 'work_email']  0.334433
## host_verifications['phone']                0.003159 **
## neighbourhoodBayside                       0.217389
## neighbourhoodBoroondara                    0.586652
## neighbourhoodBrimbank                      0.747262
## neighbourhoodCardinia                      0.000149 ***
## neighbourhoodCasey                         0.112819
## neighbourhoodDarebin                       0.580362
## neighbourhoodFrankston                     0.168113
## neighbourhoodGlen Eira                     0.095119 .
## neighbourhoodGreater Dandenong             0.708675
## neighbourhoodHobsons Bay                   0.832155
## neighbourhoodHume                          0.028395 *
## neighbourhoodKingston                      0.299452
## neighbourhoodKnox                          0.914064
## neighbourhoodManningham                    0.608801
## neighbourhoodMaribyrnong                   0.283557
```

```
## neighbourhoodMaroondah                    0.475434
## neighbourhoodMelbourne                    0.088085 .
## neighbourhoodMelton                       0.382293
## neighbourhoodMonash                       0.697830
## neighbourhoodMoonee Valley                0.283778
## neighbourhoodMoreland                     0.445028
## neighbourhoodNillumbik                    0.160412
## neighbourhoodPort Phillip                 0.734638
## neighbourhoodStonnington                  0.381617
## neighbourhoodWhitehorse                   0.968590
## neighbourhoodWhittlesea                   0.870766
## neighbourhoodWyndham                      0.559508
## neighbourhoodYarra                        0.241652
## neighbourhoodYarra Ranges                 0.026879 *
## room_typeHotel room                       0.948754
## room_typePrivate room                     9.16e-09 ***
## room_typeShared room                      0.232684
## beds                                      1.47e-09 ***
## price                                      < 2e-16 ***
## maximum_nights                            2.16e-08 ***
## number_of_reviews_ltm                     6.88e-12 ***
## number_of_reviews_l30d                     < 2e-16 ***
## instant_bookablet                         2.98e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 8027.3  on 6291  degrees of freedom
## Residual deviance: 7092.1  on 6247  degrees of freedom
## AIC: 7182.1
##
## Number of Fisher Scoring iterations: 12
```

Model3: significant LASSO plus extra variables

```
result2<-glm(HAS ~ host_identity_verified+host_acceptance_rate+host_has_profile_pic+host_response_rate+
                    room_type *instant_bookable+
             host_verifications+neighbourhood+distance_group+instant_bookable+number_of_reviews_ltm+numb
summary(result2)
```

```
##
## Call:
## glm(formula = HAS ~ host_identity_verified + host_acceptance_rate +
##     host_has_profile_pic + host_response_rate + minimum_nights_avg_ntm +
##     room_type + bedroom_per_person + bathroom_per_person + host_is_superhost +
##     availability_365 + host_total_listings_count + host_since_years +
##     log(price + 1) + maximum_nights_avg_ntm + room_type * instant_bookable +
##     host_verifications + neighbourhood + distance_group + instant_bookable +
##     number_of_reviews_ltm + number_of_reviews_l30d, family = "binomial",
##     data = train_clean)
##
```

```
## Coefficients:
##                                          Estimate Std. Error z value
## (Intercept)                            -2.934e+00  6.031e-01  -4.865
## host_identity_verifiedt                -4.278e-01  1.629e-01  -2.625
## host_acceptance_rate                   -9.881e-03  2.044e-03  -4.835
## host_has_profile_pict                  -4.567e-01  2.181e-01  -2.095
## host_response_rate                      7.223e-03  2.791e-03   2.588
## minimum_nights_avg_ntm                 -8.821e-03  5.161e-03  -1.709
## room_typeHotel room                    -1.320e+01  3.783e+02  -0.035
## room_typePrivate room                   8.967e-01  1.071e-01   8.370
## room_typeShared room                   -1.424e-01  4.366e-01  -0.326
## bedroom_per_person                      1.610e-01  1.374e-01   1.172
## bathroom_per_person                     1.488e-02  1.040e-01   0.143
## host_is_superhostt                      1.015e+00  6.736e-02  15.071
## availability_365                       -5.356e-04  2.601e-04  -2.060
## host_total_listings_count              -5.505e-03  7.637e-04  -7.208
## host_since_years                        1.577e-02  9.911e-03   1.591
## log(price + 1)                          5.474e-01  6.442e-02   8.497
## maximum_nights_avg_ntm                 -2.238e-04  6.484e-05  -3.451
## instant_bookablet                      -1.794e-01  7.841e-02  -2.288
## host_verifications['email', 'phone']    4.096e-01  8.941e-02   4.581
## host_verifications['email']            -1.198e+01  2.126e+02  -0.056
## host_verifications['phone', 'work_email'] -4.538e-01  4.103e-01  -1.106
## host_verifications['phone']             2.606e-01  1.332e-01   1.956
## neighbourhoodBayside                    3.821e-01  3.507e-01   1.090
## neighbourhoodBoroondara                -1.096e-01  3.305e-01  -0.332
## neighbourhoodBrimbank                   1.815e-01  3.559e-01   0.510
## neighbourhoodCardinia                   1.528e+00  4.028e-01   3.793
## neighbourhoodCasey                      4.503e-01  3.481e-01   1.294
## neighbourhoodDarebin                   -5.943e-02  3.390e-01  -0.175
## neighbourhoodFrankston                  5.062e-01  3.410e-01   1.485
## neighbourhoodGlen Eira                  4.760e-01  3.172e-01   1.501
## neighbourhoodGreater Dandenong         -4.595e-02  4.362e-01  -0.105
## neighbourhoodHobsons Bay               -1.105e-01  3.667e-01  -0.301
## neighbourhoodHume                       7.584e-01  3.643e-01   2.082
## neighbourhoodKingston                   4.675e-01  3.449e-01   1.356
## neighbourhoodKnox                       2.433e-02  4.349e-01   0.056
## neighbourhoodManningham                -3.009e-01  4.258e-01  -0.707
## neighbourhoodMaribyrnong               -5.971e-01  3.498e-01  -1.707
## neighbourhoodMaroondah                  5.262e-01  4.804e-01   1.095
## neighbourhoodMelbourne                 -6.922e-01  3.374e-01  -2.052
## neighbourhoodMelton                     2.861e-01  3.996e-01   0.716
## neighbourhoodMonash                    -9.788e-02  3.374e-01  -0.290
## neighbourhoodMoonee Valley              6.267e-02  3.572e-01   0.175
## neighbourhoodMoreland                   1.385e-01  3.226e-01   0.429
## neighbourhoodNillumbik                  6.458e-01  4.419e-01   1.461
## neighbourhoodPort Phillip              -1.834e-01  3.056e-01  -0.600
## neighbourhoodStonnington               -3.482e-02  3.157e-01  -0.110
## neighbourhoodWhitehorse                 1.184e-01  3.260e-01   0.363
## neighbourhoodWhittlesea                 1.273e-02  4.126e-01   0.031
## neighbourhoodWyndham                    1.460e-01  3.077e-01   0.475
## neighbourhoodYarra                     -7.939e-02  3.255e-01  -0.244
## neighbourhoodYarra Ranges               6.365e-01  2.857e-01   2.228
## distance_group2-5 km                    3.440e-02  1.468e-01   0.234
```

```
## distance_group5-10 km                        -1.161e-01  1.882e-01  -0.617
## distance_group>10 km                         -4.841e-01  2.428e-01  -1.994
## number_of_reviews_ltm                        -1.607e-02  2.407e-03  -6.676
## number_of_reviews_l30d                        1.609e-01  2.101e-02   7.658
## room_typeHotel room:instant_bookablet         7.845e-01  4.443e+02   0.002
## room_typePrivate room:instant_bookablet      -4.770e-01  1.889e-01  -2.525
## room_typeShared room:instant_bookablet       -2.266e-01  8.661e-01  -0.262
##                                              Pr(>|z|)
## (Intercept)                                  1.14e-06 ***
## host_identity_verifiedt                      0.008653 **
## host_acceptance_rate                         1.33e-06 ***
## host_has_profile_pict                        0.036208 *
## host_response_rate                           0.009655 **
## minimum_nights_avg_ntm                       0.087449 .
## room_typeHotel room                          0.972167
## room_typePrivate room                         < 2e-16 ***
## room_typeShared room                         0.744321
## bedroom_per_person                           0.241286
## bathroom_per_person                          0.886248
## host_is_superhostt                            < 2e-16 ***
## availability_365                             0.039445 *
## host_total_listings_count                    5.67e-13 ***
## host_since_years                             0.111607
## log(price + 1)                                < 2e-16 ***
## maximum_nights_avg_ntm                       0.000559 ***
## instant_bookablet                            0.022130 *
## host_verifications['email', 'phone']         4.63e-06 ***
## host_verifications['email']                  0.955069
## host_verifications['phone', 'work_email']    0.268715
## host_verifications['phone']                  0.050444 .
## neighbourhoodBayside                         0.275876
## neighbourhoodBoroondara                      0.740266
## neighbourhoodBrimbank                        0.610085
## neighbourhoodCardinia                        0.000149 ***
## neighbourhoodCasey                           0.195749
## neighbourhoodDarebin                         0.860850
## neighbourhoodFrankston                       0.137670
## neighbourhoodGlen Eira                       0.133447
## neighbourhoodGreater Dandenong               0.916098
## neighbourhoodHobsons Bay                     0.763259
## neighbourhoodHume                            0.037376 *
## neighbourhoodKingston                        0.175253
## neighbourhoodKnox                            0.955381
## neighbourhoodManningham                      0.479812
## neighbourhoodMaribyrnong                     0.087870 .
## neighbourhoodMaroondah                       0.273424
## neighbourhoodMelbourne                       0.040209 *
## neighbourhoodMelton                          0.473901
## neighbourhoodMonash                          0.771699
## neighbourhoodMoonee Valley                   0.860727
## neighbourhoodMoreland                        0.667648
## neighbourhoodNillumbik                       0.143882
## neighbourhoodPort Phillip                    0.548506
## neighbourhoodStonnington                     0.912178
```

```
## neighbourhoodWhitehorse                    0.716493
## neighbourhoodWhittlesea                    0.975392
## neighbourhoodWyndham                       0.635062
## neighbourhoodYarra                         0.807317
## neighbourhoodYarra Ranges                  0.025874 *
## distance_group2-5 km                       0.814718
## distance_group5-10 km                      0.537367
## distance_group>10 km                       0.046176 *
## number_of_reviews_ltm                      2.45e-11 ***
## number_of_reviews_l30d                     1.88e-14 ***
## room_typeHotel room:instant_bookablet      0.998591
## room_typePrivate room:instant_bookablet    0.011562 *
## room_typeShared room:instant_bookablet     0.793644
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 8027.3  on 6291  degrees of freedom
## Residual deviance: 6989.4  on 6233  degrees of freedom
## AIC: 7107.4
##
## Number of Fisher Scoring iterations: 12
```

```
p2 <- predict(result2, newdata = test_clean, type = "response")
pc2 <- ifelse(p2 > 0.5, 1, 0)
confusion_matrix_2 <- table(Predicted = pc2, Actual = test_clean$HAS)
print(confusion_matrix_2)
```

```
##          Actual
## Predicted    0    1
##         0 1566  577
##         1  227  328
```

```
accuracy_2 <- mean(pc2 == test_clean$HAS)
print(paste("Accuracy:", round(accuracy_2, 4)))
```

```
## [1] "Accuracy: 0.702"
```

```
# ROC
roc <- roc(response = test_clean$HAS, predictor = p2)
```

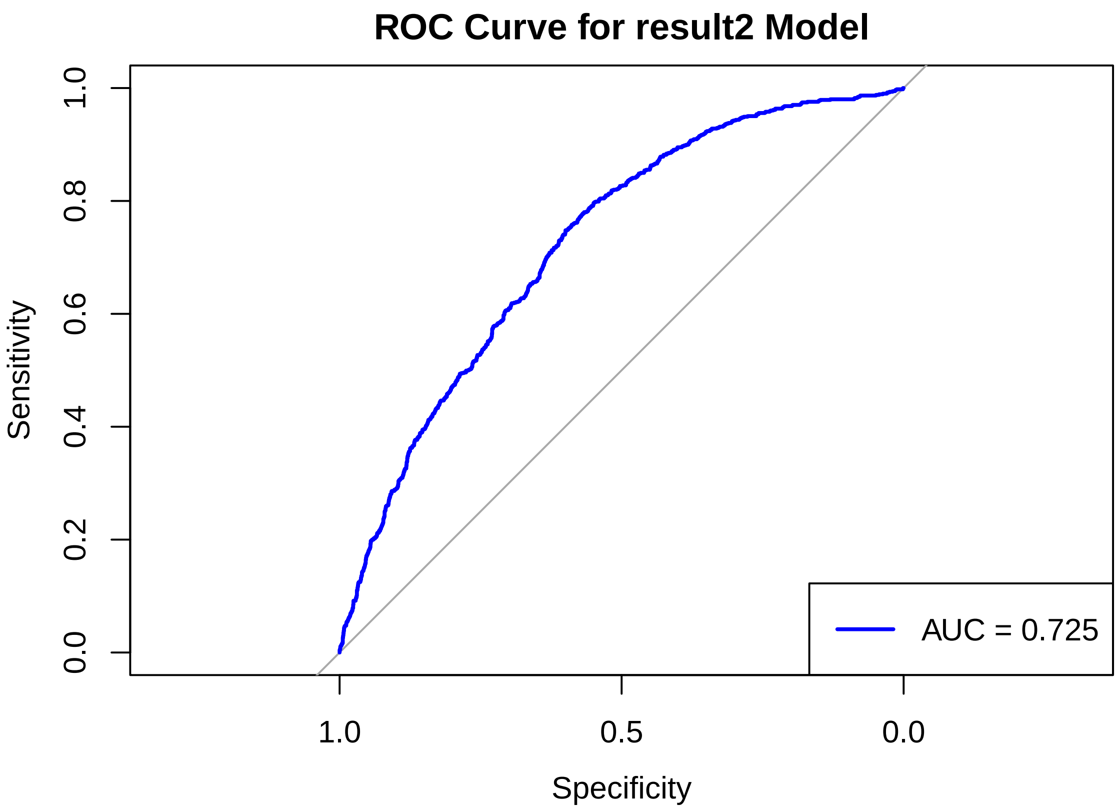
```
## Setting levels: control = 0, case = 1
```

```
## Setting direction: controls < cases
```

```
plot(roc, col = "blue", lwd = 2, main = "ROC Curve for result2 Model")

# AUC
auc2 <- auc(roc)
print(paste("AUC:", round(auc2, 4)))
```

```
## [1] "AUC: 0.725"
```

```r
legend("bottomright", legend = paste("AUC =", round(auc2, 4)), col = "blue", lwd = 2)
```




```r
vif(result2)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

```
##                                 GVIF Df GVIF^(1/(2*Df))
## host_identity_verified      1.037548  1        1.018601
## host_acceptance_rate        1.747691  1        1.322003
## host_has_profile_pic        1.083341  1        1.040837
## host_response_rate          1.479441  1        1.216323
## minimum_nights_avg_ntm      1.076751  1        1.037666
## room_type                  11.456708  3        1.501447
## bedroom_per_person          1.341892  1        1.158401
## bathroom_per_person         1.652653  1        1.285555
## host_is_superhost           1.346416  1        1.160352
## availability_365            1.144453  1        1.069791
## host_total_listings_count   1.284530  1        1.133371
## host_since_years            1.301567  1        1.140862
## log(price + 1)              2.103415  1        1.450315
## maximum_nights_avg_ntm      1.117296  1        1.057022
## instant_bookable            1.357480  1        1.165109
```

```
## host_verifications          1.272135  4        1.030544
## neighbourhood              71.464479 29        1.076384
## distance_group             49.263026  3        1.914639
## number_of_reviews_ltm       2.203756  1        1.484505
## number_of_reviews_l30d      1.930513  1        1.389429
## room_type:instant_bookable  6.817591  3        1.377014
```

Question 2

diagnostic tools

```r
#cross-validation
library(boot)
```

```
##
## Attaching package: 'boot'

## The following object is masked from 'package:car':
##
##     logit
```

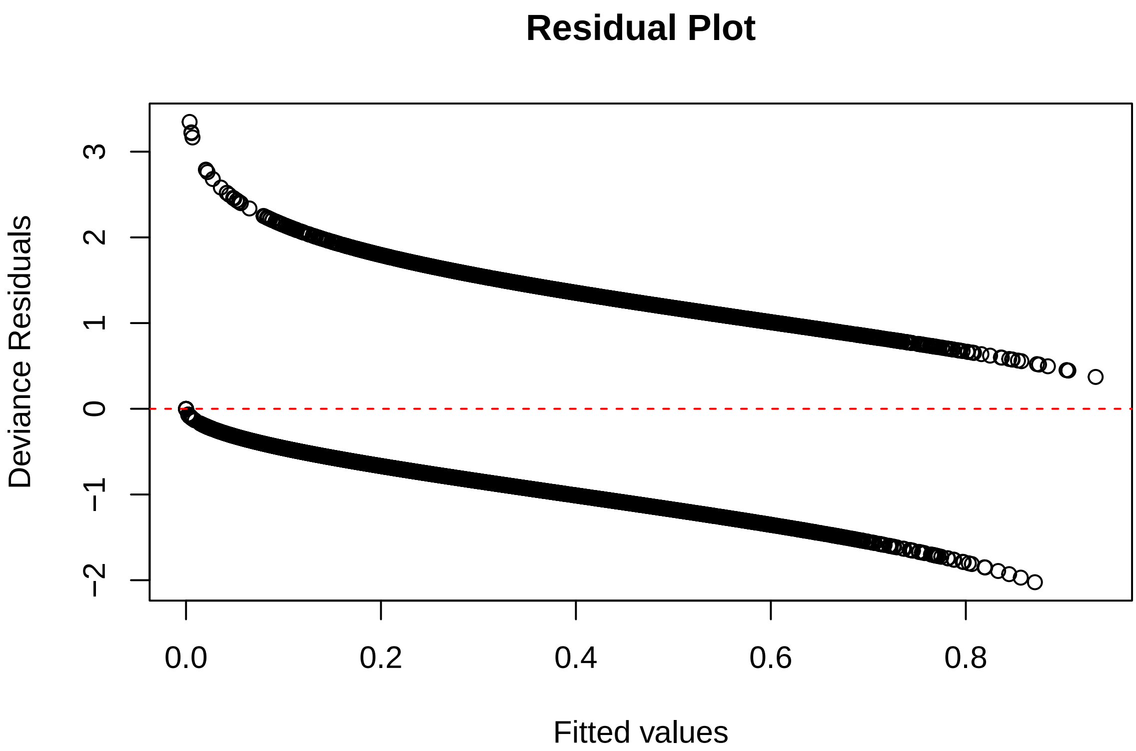
```r
cv_result_2 <- cv.glm(train_clean, result2, K = 10)
cv_result_2$delta
```

```
## [1] 0.1922901 0.1920898
```

```r
pred_prob <- predict(lasso_cv, newx = x_test, s = "lambda.min", type = "response")
y_test_numeric <- as.numeric(as.character(y_test))
test_mse <- mean((pred_prob - y_test_numeric)^2)
test_mse
```

```
## [1] 0.1896758
```

```r
#residual plots
# logistic model is named `result2`
residuals <- residuals(result2, type = "deviance")
fitted_vals <- fitted(result2)

plot(fitted_vals, residuals,
     xlab = "Fitted values", ylab = "Deviance Residuals",
     main = "Residual Plot")
abline(h = 0, col = "red", lty = 2)
```



Question 4 Prediction model for price

data cleaning and spliting

```r
#data import from mel
review_cols <- mel_original[, grepl("review_scores", names(mel_original))]
mel2 <- cbind(mel, review_cols)
str(mel2)
```

```
## 'data.frame':    8990 obs. of  51 variables:
##  $ host_since                : Date, format: "2022-02-24" "2020-12-30" ...
##  $ host_response_time        : Factor w/ 4 levels "a few days or more",..: 4 3 4 4 4 3 4 4 4 3 ...
##  $ host_response_rate        : num  95 89 100 100 97 82 100 100 100 100 ...
##  $ host_acceptance_rate      : num  99 57 99 100 94 41 100 100 100 56 ...
##  $ host_is_superhost         : Factor w/ 2 levels "f","t": 1 1 1 1 1 1 1 1 2 1 ...
##  $ host_listings_count       : int  8 3 109 1 7 346 2 1 4 1 ...
##  $ host_total_listings_count: int  8 4 270 2 7 368 3 1 6 3 ...
##  $ host_verifications        : Factor w/ 5 levels "['email', 'phone', 'work_email']",..: 2 5 1 1 2 1
##  $ host_has_profile_pic      : Factor w/ 2 levels "f","t": 2 2 2 2 2 2 2 2 2 2 ...
##  $ host_identity_verified    : Factor w/ 2 levels "f","t": 2 2 2 2 2 2 2 2 2 2 ...
##  $ neighbourhood             : Factor w/ 30 levels "Banyule","Bayside",..: 24 4 18 9 24 18 29 18 29 9
##  $ latitude                  : num  -37.9 -37.8 -37.8 -37.9 -37.8 ...
##  $ longitude                 : num  145 145 145 145 145 ...
```

```
##  $ property_type              : Factor w/ 72 levels "Barn","Boat",..: 63 39 19 39 19 39 39 21 19 44 ..
##  $ room_type                  : Factor w/ 4 levels "Entire home/apt",..: 4 3 1 3 1 3 3 1 1 3 ...
##  $ accommodates               : int  1 1 3 1 2 2 2 3 4 2 ...
##  $ bathrooms                  : num  1 1 1 1 1 2 1 1 1 1 ...
##  $ bathrooms_text             : Factor w/ 34 levels "","0 baths","0 shared baths",..: 6 6 4 4 4 10 5 4
##  $ bedrooms                   : int  1 1 2 1 1 1 1 1 1 1 ...
##  $ beds                       : int  1 0 2 1 1 2 2 2 2 0 ...
##  $ amenities                  : Factor w/ 8669 levels "[\"55\\\" HDTV with Amazon Prime Video, Disney+
##  $ price                      : num  123 50 166 120 202 80 75 104 175 55 ...
##  $ minimum_nights             : int  1 1 3 1 2 7 1 5 2 2 ...
##  $ maximum_nights             : int  365 365 90 365 365 365 1125 365 365 1125 ...
##  $ minimum_minimum_nights     : int  1 1 3 1 2 7 1 2 3 2 ...
##  $ maximum_minimum_nights     : int  1 1 3 1 2 7 1 5 3 2 ...
##  $ minimum_maximum_nights     : int  30 365 1125 365 365 365 1125 365 365 1125 ...
##  $ maximum_maximum_nights     : int  1125 365 1125 365 365 365 1125 365 365 1125 ...
##  $ minimum_nights_avg_ntm     : num  1 1 3 1 2 7 1 5 3 2 ...
##  $ maximum_nights_avg_ntm     : num  262 365 1125 365 365 ...
##  $ has_availability           : Factor w/ 1 level "t": 1 1 1 1 1 1 1 1 1 1 ...
##  $ availability_30            : int  29 24 12 29 0 4 6 0 13 20 ...
##  $ availability_60            : int  59 54 42 59 0 34 24 0 13 50 ...
##  $ availability_90            : int  89 84 72 89 0 64 53 20 13 80 ...
##  $ availability_365           : int  352 359 88 269 0 64 233 215 13 355 ...
##  $ number_of_reviews          : int  30 5 2 1 2 2 100 5 19 11 ...
##  $ number_of_reviews_ltm      : int  9 5 2 1 2 2 15 5 19 6 ...
##  $ number_of_reviews_l30d     : int  1 1 1 1 2 0 6 3 0 1 ...
##  $ first_review               : Factor w/ 2431 levels "2010/11/24","2010/12/5",..: 1814 2332 2365 2429
##  $ last_review                : Factor w/ 654 levels "2014/3/10","2016/12/3",..: 616 616 626 652 637 5
##  $ instant_bookable           : Factor w/ 2 levels "f","t": 2 1 1 1 1 1 2 2 2 1 ...
##  $ reviews_per_month          : num  1.21 0.81 0.98 1 2 0.17 1.08 3.85 1.75 0.68 ...
##  $ X                          : logi  NA NA NA NA NA NA ...
##  $ X.1                        : logi  NA NA NA NA NA NA ...
##  $ HAS                        : num  0 0 0 1 0 0 0 1 0 1 ...
##  $ distance_meters            : num  5444 10721 1186 15217 2404 ...
##  $ distance_km                : num  5.44 10.72 1.19 15.22 2.4 ...
##  $ distance_group             : Factor w/ 4 levels "0-2 km","2-5 km",..: 3 4 1 4 2 1 2 2 1 4 ...
##  $ host_since_years           : num  3.43 4.58 10.78 11.16 1.61 ...
##  $ bathroom_per_person        : num  1 1 0.333 1 0.5 ...
##  $ bedroom_per_person         : num  1 1 0.667 1 0.5 ...
```

```r
drop_cols <- c("amenities", "bathrooms_text",
               "has_availability", "host_since","property_type","first_review","last_review","X","X.1")
mel2_f <- mel[, !(names(mel2) %in% drop_cols)] #mel2_final data
str(mel2_f)
```

```
## 'data.frame':    8990 obs. of  42 variables:
##  $ host_response_time         : Factor w/ 4 levels "a few days or more",..: 4 3 4 4 4 3 4 4 4 3 ...
##  $ host_response_rate         : num  95 89 100 100 97 82 100 100 100 100 ...
##  $ host_acceptance_rate       : num  99 57 99 100 94 41 100 100 100 56 ...
##  $ host_is_superhost          : Factor w/ 2 levels "f","t": 1 1 1 1 1 1 1 1 2 1 ...
##  $ host_listings_count        : int  8 3 109 1 7 346 2 1 4 1 ...
##  $ host_total_listings_count: int  8 4 270 2 7 368 3 1 6 3 ...
##  $ host_verifications         : Factor w/ 5 levels "['email', 'phone', 'work_email']",..: 2 5 1 1 2 1
##  $ host_has_profile_pic       : Factor w/ 2 levels "f","t": 2 2 2 2 2 2 2 2 2 2 ...
##  $ host_identity_verified     : Factor w/ 2 levels "f","t": 2 2 2 2 2 2 2 2 2 2 ...
```

```
## $ neighbourhood            : Factor w/ 30 levels "Banyule","Bayside",..: 24 4 18 9 24 18 29 18 29 9
## $ latitude                 : num  -37.9 -37.8 -37.8 -37.9 -37.8 ...
## $ longitude                : num  145 145 145 145 145 ...
## $ room_type                : Factor w/ 4 levels "Entire home/apt",..: 4 3 1 3 1 3 3 1 1 3 ...
## $ accommodates             : int  1 1 3 1 2 2 2 3 4 2 ...
## $ bathrooms                : num  1 1 1 1 1 2 1 1 1 1 ...
## $ bedrooms                 : int  1 1 2 1 1 1 1 1 1 1 ...
## $ beds                     : int  1 0 2 1 1 2 2 2 2 0 ...
## $ price                    : num  123 50 166 120 202 80 75 104 175 55 ...
## $ minimum_nights           : int  1 1 3 1 2 7 1 5 2 2 ...
## $ maximum_nights           : int  365 365 90 365 365 365 1125 365 365 1125 ...
## $ minimum_minimum_nights   : int  1 1 3 1 2 7 1 2 3 2 ...
## $ maximum_minimum_nights   : int  1 1 3 1 2 7 1 5 3 2 ...
## $ minimum_maximum_nights   : int  30 365 1125 365 365 365 1125 365 365 1125 ...
## $ maximum_maximum_nights   : int  1125 365 1125 365 365 365 1125 365 365 1125 ...
## $ minimum_nights_avg_ntm   : num  1 1 3 1 2 7 1 5 3 2 ...
## $ maximum_nights_avg_ntm   : num  262 365 1125 365 365 ...
## $ availability_30          : int  29 24 12 29 0 4 6 0 13 20 ...
## $ availability_60          : int  59 54 42 59 0 34 24 0 13 50 ...
## $ availability_90          : int  89 84 72 89 0 64 53 20 13 80 ...
## $ availability_365         : int  352 359 88 269 0 64 233 215 13 355 ...
## $ number_of_reviews        : int  30 5 2 1 2 2 100 5 19 11 ...
## $ number_of_reviews_ltm    : int  9 5 2 1 2 2 15 5 19 6 ...
## $ number_of_reviews_l30d   : int  1 1 1 1 2 0 6 3 0 1 ...
## $ instant_bookable         : Factor w/ 2 levels "f","t": 2 1 1 1 1 1 2 2 2 1 ...
## $ reviews_per_month        : num  1.21 0.81 0.98 1 2 0.17 1.08 3.85 1.75 0.68 ...
## $ HAS                      : num  0 0 0 1 0 0 0 1 0 1 ...
## $ distance_meters          : num  5444 10721 1186 15217 2404 ...
## $ distance_km              : num  5.44 10.72 1.19 15.22 2.4 ...
## $ distance_group           : Factor w/ 4 levels "0-2 km","2-5 km",..: 3 4 1 4 2 1 2 2 1 4 ...
## $ host_since_years         : num  3.43 4.58 10.78 11.16 1.61 ...
## $ bathroom_per_person      : num  1 1 0.333 1 0.5 ...
## $ bedroom_per_person       : num  1 1 0.667 1 0.5 ...
```

```r
#split data into training and test sets
# Load required libraries
mel2_f$LogPrice <- log(mel2_f$price + 1)

library(glmnet)
set.seed(123)

n <- nrow(mel2_f)
train_idx <- sample(seq_len(n), size = 0.8 * n)

train2 <- mel2_f[train_idx, ]
test2 <- mel2_f[-train_idx, ]

# Convert categorical vars to dummies
x_train2 <- model.matrix(LogPrice ~ . - price, data = train2)[, -1]
y_train2 <- train2$LogPrice
x_test2 <- model.matrix(LogPrice ~ . - price, data = test2)[, -1]
y_test2 <- test2$LogPrice
```

LASSO selection

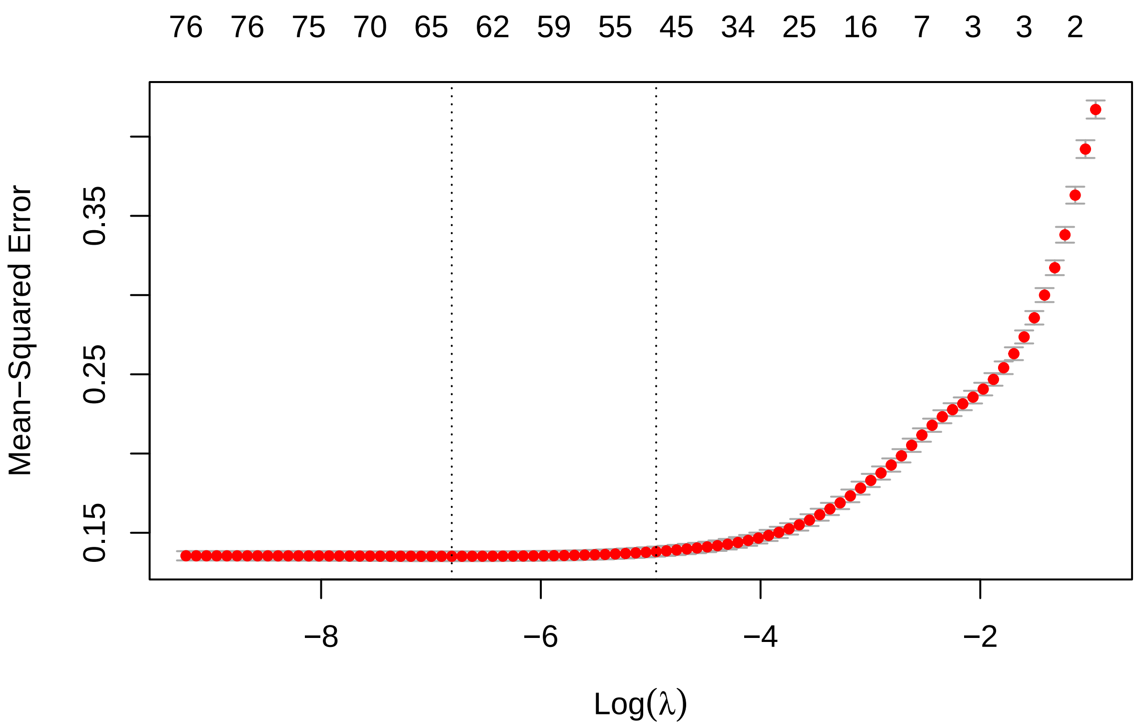
```r
# Fit LASSO
lasso_cv2 <- cv.glmnet(x_train2, y_train2, alpha = 1)
best_lambda2 <- lasso_cv2$lambda.min

# Predict on test data
pred2 <- predict(lasso_cv2, s = best_lambda2, newx = x_test2)
pred2_L <- exp(pred2) - 1
actual_price <- test2$price


# Compute test MSE
test_mse2_L <- mean((pred2_L - actual_price)^2)
rmse_L <- sqrt(test_mse2_L)
print(paste("Test MSE for LASSO price:", round(test_mse2_L, 4)))
```

```
## [1] "Test MSE for LASSO price: 10464.4273"
```

```r
print(paste("Test RMSE for LASSO price:", round(rmse_L, 4)))
```

```
## [1] "Test RMSE for LASSO price: 102.2958"
```

```r
# Optional: plot coefficient path
plot(lasso_cv2)
```



```r
#lasso coefficient
lasso_coef2 <- coef(lasso_cv2, s = "lambda.min")
print(lasso_coef2)
```

```
## 79 x 1 sparse Matrix of class "dgCMatrix"
##                                                     s1
## (Intercept)                              -6.380027e+00
## host_response_timewithin a day            5.475360e-02
## host_response_timewithin a few hours      .
## host_response_timewithin an hour          6.356542e-02
## host_response_rate                        .
## host_acceptance_rate                      6.658070e-04
## host_is_superhostt                        .
## host_listings_count                      -1.138441e-03
## host_total_listings_count                 3.843314e-04
## host_verifications['email', 'phone']     -2.655702e-02
## host_verifications['email']               .
## host_verifications['phone', 'work_email'] .
## host_verifications['phone']              -5.145771e-02
## host_has_profile_pict                     3.684410e-02
## host_identity_verifiedt                  -1.173141e-02
## neighbourhoodBayside                      2.735226e-01
## neighbourhoodBoroondara                  -1.793278e-03
## neighbourhoodBrimbank                    -1.392909e-01
## neighbourhoodCardinia                     1.544702e-01
## neighbourhoodCasey                       -1.109010e-01
## neighbourhoodDarebin                     -1.069169e-01
## neighbourhoodFrankston                    .
## neighbourhoodGlen Eira                    1.222798e-02
## neighbourhoodGreater Dandenong           -1.570854e-01
## neighbourhoodHobsons Bay                 -7.870917e-02
## neighbourhoodHume                         .
## neighbourhoodKingston                     1.827784e-01
## neighbourhoodKnox                        -1.006087e-01
## neighbourhoodManningham                  -1.974135e-02
## neighbourhoodMaribyrnong                 -1.975102e-01
## neighbourhoodMaroondah                   -4.648603e-02
## neighbourhoodMelbourne                    1.378507e-01
## neighbourhoodMelton                      -3.740013e-02
## neighbourhoodMonash                       .
## neighbourhoodMoonee Valley                2.469536e-02
## neighbourhoodMoreland                    -4.309433e-02
## neighbourhoodNillumbik                    2.273880e-01
## neighbourhoodPort Phillip                 2.195879e-01
## neighbourhoodStonnington                  5.644189e-02
## neighbourhoodWhitehorse                  -5.894296e-02
## neighbourhoodWhittlesea                  -4.964003e-02
## neighbourhoodWyndham                     -2.292651e-01
## neighbourhoodYarra                        1.405312e-01
## neighbourhoodYarra Ranges                 4.987566e-01
## latitude                                 -1.977577e-01
## longitude                                 2.563260e-02
## room_typeHotel room                       2.586770e-01
```

```
## room_typePrivate room                      -5.089357e-01
## room_typeShared room                       -9.267587e-01
## accommodates                                .
## bathrooms                                   1.245582e-01
## bedrooms                                    2.186650e-01
## beds                                        1.188444e-03
## minimum_nights                             -9.988671e-04
## maximum_nights                             -7.877030e-06
## minimum_minimum_nights                     -1.705997e-03
## maximum_minimum_nights                      1.492757e-04
## minimum_maximum_nights                      .
## maximum_maximum_nights                      .
## minimum_nights_avg_ntm                      .
## maximum_nights_avg_ntm                      .
## availability_30                             1.156878e-02
## availability_60                            -1.125812e-04
## availability_90                            -1.695389e-03
## availability_365                            5.387042e-04
## number_of_reviews                          -3.905387e-04
## number_of_reviews_ltm                       .
## number_of_reviews_l30d                     -2.727503e-02
## instant_bookablet                           8.381798e-02
## reviews_per_month                          -5.090886e-03
## HAS                                         1.080986e-01
## distance_meters                            -3.686688e-06
## distance_km                                 .
## distance_group2-5 km                        .
## distance_group5-10 km                      -6.141527e-02
## distance_group>10 km                       -2.356326e-01
## host_since_years                            3.499474e-04
## bathroom_per_person                        -1.640527e-01
## bedroom_per_person                         -2.680857e-01
```

```r
# Extract the names of variables selected by LASSO
selected_vars <- rownames(lasso_coef2)[which(lasso_coef2 != 0)]
selected_vars2 <- selected_vars[selected_vars != "(Intercept)"]  # exclude intercept

x_train_selected2 <- x_train2[, selected_vars2]
Price_lasso<- lm(y_train2 ~ ., data = as.data.frame(x_train_selected2))


# View summary with p-values and coefficients later if lasso win'summary(Price_lasso)'
```

Stepwise selection

```r
#stepwise model

# Drop 'price' column from predictors before fitting the model
train2_noprice <- train2[, !names(train2) %in% "price"]

# Fit full linear model without 'price' as predictor
full_model2 <- lm(LogPrice ~ ., data = train2_noprice)
```

```r
# Perform stepwise selection
step_model2 <- step(full_model2, direction = "both")
```

```
## Start:  AIC=-14405.8
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_is_superhost + host_listings_count + host_total_listings_count +
##     host_verifications + host_has_profile_pic + host_identity_verified +
##     neighbourhood + latitude + longitude + room_type + accommodates +
##     bathrooms + bedrooms + beds + minimum_nights + maximum_nights +
##     minimum_minimum_nights + maximum_minimum_nights + minimum_maximum_nights +
##     maximum_maximum_nights + minimum_nights_avg_ntm + maximum_nights_avg_ntm +
##     availability_30 + availability_60 + availability_90 + availability_365 +
##     number_of_reviews + number_of_reviews_ltm + number_of_reviews_l30d +
##     instant_bookable + reviews_per_month + HAS + distance_meters +
##     distance_km + distance_group + host_since_years + bathroom_per_person +
##     bedroom_per_person
##
##
## Step:  AIC=-14405.8
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_is_superhost + host_listings_count + host_total_listings_count +
##     host_verifications + host_has_profile_pic + host_identity_verified +
##     neighbourhood + latitude + longitude + room_type + accommodates +
##     bathrooms + bedrooms + beds + minimum_nights + maximum_nights +
##     minimum_minimum_nights + maximum_minimum_nights + minimum_maximum_nights +
##     maximum_maximum_nights + minimum_nights_avg_ntm + maximum_nights_avg_ntm +
##     availability_30 + availability_60 + availability_90 + availability_365 +
##     number_of_reviews + number_of_reviews_ltm + number_of_reviews_l30d +
##     instant_bookable + reviews_per_month + HAS + distance_meters +
##     distance_group + host_since_years + bathroom_per_person +
##     bedroom_per_person
##
##                               Df Sum of Sq    RSS    AIC
## - maximum_minimum_nights      1     0.000 949.56 -14408
## - host_since_years            1     0.008 949.57 -14408
## - host_is_superhost           1     0.011 949.57 -14408
## - minimum_nights_avg_ntm      1     0.012 949.58 -14408
## - maximum_nights_avg_ntm      1     0.014 949.58 -14408
## - minimum_maximum_nights      1     0.021 949.58 -14408
## - number_of_reviews_ltm       1     0.036 949.60 -14408
## - beds                        1     0.083 949.65 -14407
## - host_identity_verified      1     0.084 949.65 -14407
## - maximum_maximum_nights      1     0.137 949.70 -14407
## - accommodates                1     0.138 949.70 -14407
## - latitude                    1     0.158 949.72 -14407
## - availability_60             1     0.174 949.74 -14406
## - maximum_nights              1     0.209 949.77 -14406
## - reviews_per_month           1     0.226 949.79 -14406
## - host_has_profile_pic        1     0.231 949.79 -14406
## - longitude                   1     0.249 949.81 -14406
## - minimum_minimum_nights      1     0.260 949.82 -14406
## <none>                                    949.56 -14406
## - minimum_nights              1     0.270 949.83 -14406
```

```
## - host_acceptance_rate       1     0.531  950.09 -14404
## - host_response_rate         1     0.534  950.10 -14404
## - availability_90            1     0.551  950.11 -14404
## - host_verifications         4     1.742  951.31 -14401
## - distance_meters            1     1.889  951.45 -14394
## - host_response_time         3     3.401  952.96 -14386
## - number_of_reviews          1     3.351  952.91 -14382
## - host_total_listings_count  1     4.073  953.64 -14377
## - distance_group             3     5.726  955.29 -14369
## - host_listings_count        1     8.233  957.80 -14346
## - instant_bookable           1     8.328  957.89 -14345
## - number_of_reviews_l30d     1     8.968  958.53 -14340
## - bathroom_per_person        1     9.813  959.38 -14334
## - bedroom_per_person         1    10.813  960.38 -14326
## - bathrooms                  1    15.082  964.64 -14294
## - availability_30            1    15.367  964.93 -14292
## - HAS                        1    16.981  966.54 -14280
## - availability_365           1    19.547  969.11 -14261
## - bedrooms                   1    39.033  988.60 -14118
## - neighbourhood             29    88.205 1037.77 -13825
## - room_type                  3   198.280 1147.84 -13048
##
## Step:  AIC=-14407.8
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_is_superhost + host_listings_count + host_total_listings_count +
##     host_verifications + host_has_profile_pic + host_identity_verified +
##     neighbourhood + latitude + longitude + room_type + accommodates +
##     bathrooms + bedrooms + beds + minimum_nights + maximum_nights +
##     minimum_minimum_nights + minimum_maximum_nights + maximum_maximum_nights +
##     minimum_nights_avg_ntm + maximum_nights_avg_ntm + availability_30 +
##     availability_60 + availability_90 + availability_365 + number_of_reviews +
##     number_of_reviews_ltm + number_of_reviews_l30d + instant_bookable +
##     reviews_per_month + HAS + distance_meters + distance_group +
##     host_since_years + bathroom_per_person + bedroom_per_person
##
##                            Df Sum of Sq     RSS    AIC
## - host_since_years          1     0.008  949.57 -14410
## - host_is_superhost         1     0.011  949.57 -14410
## - maximum_nights_avg_ntm    1     0.015  949.58 -14410
## - minimum_maximum_nights    1     0.021  949.58 -14410
## - number_of_reviews_ltm     1     0.036  949.60 -14410
## - beds                      1     0.083  949.65 -14409
## - host_identity_verified    1     0.085  949.65 -14409
## - minimum_nights_avg_ntm    1     0.099  949.66 -14409
## - maximum_maximum_nights    1     0.138  949.70 -14409
## - accommodates              1     0.138  949.70 -14409
## - latitude                  1     0.158  949.72 -14409
## - availability_60           1     0.174  949.74 -14408
## - maximum_nights            1     0.209  949.77 -14408
## - reviews_per_month         1     0.226  949.79 -14408
## - host_has_profile_pic      1     0.231  949.79 -14408
## - longitude                 1     0.249  949.81 -14408
## <none>                                   949.56 -14408
## - minimum_nights            1     0.280  949.84 -14408
```

```
## - minimum_minimum_nights     1     0.489  950.05 -14406
## + maximum_minimum_nights     1     0.000  949.56 -14406
## - host_acceptance_rate       1     0.531  950.10 -14406
## - host_response_rate         1     0.535  950.10 -14406
## - availability_90            1     0.551  950.12 -14406
## - host_verifications         4     1.743  951.31 -14403
## - distance_meters            1     1.888  951.45 -14396
## - host_response_time         3     3.401  952.97 -14388
## - number_of_reviews          1     3.351  952.91 -14384
## - host_total_listings_count  1     4.076  953.64 -14379
## - distance_group             3     5.734  955.30 -14370
## - host_listings_count        1     8.237  957.80 -14348
## - instant_bookable           1     8.352  957.92 -14347
## - number_of_reviews_l30d     1     8.970  958.53 -14342
## - bathroom_per_person        1     9.813  959.38 -14336
## - bedroom_per_person         1    10.813  960.38 -14328
## - bathrooms                  1    15.081  964.65 -14296
## - availability_30            1    15.370  964.93 -14294
## - HAS                        1    16.991  966.55 -14282
## - availability_365           1    19.629  969.19 -14263
## - bedrooms                   1    39.033  988.60 -14120
## - neighbourhood             29    88.213 1037.78 -13827
## - room_type                  3   198.316 1147.88 -13050
##
## Step:  AIC=-14409.74
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_is_superhost + host_listings_count + host_total_listings_count +
##     host_verifications + host_has_profile_pic + host_identity_verified +
##     neighbourhood + latitude + longitude + room_type + accommodates +
##     bathrooms + bedrooms + beds + minimum_nights + maximum_nights +
##     minimum_minimum_nights + minimum_maximum_nights + maximum_maximum_nights +
##     minimum_nights_avg_ntm + maximum_nights_avg_ntm + availability_30 +
##     availability_60 + availability_90 + availability_365 + number_of_reviews +
##     number_of_reviews_ltm + number_of_reviews_l30d + instant_bookable +
##     reviews_per_month + HAS + distance_meters + distance_group +
##     bathroom_per_person + bedroom_per_person
##
##                             Df Sum of Sq     RSS    AIC
## - host_is_superhost          1     0.013  949.58 -14412
## - maximum_nights_avg_ntm     1     0.016  949.59 -14412
## - minimum_maximum_nights     1     0.021  949.59 -14412
## - number_of_reviews_ltm      1     0.035  949.61 -14412
## - beds                       1     0.084  949.66 -14411
## - host_identity_verified     1     0.085  949.66 -14411
## - minimum_nights_avg_ntm     1     0.100  949.67 -14411
## - accommodates               1     0.139  949.71 -14411
## - maximum_maximum_nights     1     0.144  949.72 -14411
## - latitude                   1     0.159  949.73 -14410
## - availability_60            1     0.174  949.75 -14410
## - maximum_nights             1     0.207  949.78 -14410
## - reviews_per_month          1     0.244  949.82 -14410
## - host_has_profile_pic       1     0.249  949.82 -14410
## - longitude                  1     0.250  949.82 -14410
## <none>                                    949.57 -14410
```

```
## - minimum_nights             1     0.279  949.85 -14410
## - minimum_minimum_nights     1     0.487  950.06 -14408
## + host_since_years           1     0.008  949.56 -14408
## - host_acceptance_rate       1     0.525  950.10 -14408
## + maximum_minimum_nights     1     0.000  949.57 -14408
## - host_response_rate         1     0.528  950.10 -14408
## - availability_90            1     0.555  950.13 -14408
## - host_verifications         4     1.940  951.51 -14403
## - distance_meters            1     1.889  951.46 -14397
## - host_response_time         3     3.395  952.97 -14390
## - number_of_reviews          1     3.473  953.04 -14386
## - host_total_listings_count  1     4.101  953.67 -14381
## - distance_group             3     5.764  955.34 -14372
## - host_listings_count        1     8.271  957.84 -14349
## - instant_bookable           1     8.362  957.93 -14349
## - number_of_reviews_l30d     1     8.962  958.53 -14344
## - bathroom_per_person        1     9.809  959.38 -14338
## - bedroom_per_person         1    10.818  960.39 -14330
## - bathrooms                  1    15.075  964.65 -14298
## - availability_30            1    15.364  964.94 -14296
## - HAS                        1    17.059  966.63 -14284
## - availability_365           1    19.639  969.21 -14264
## - bedrooms                   1    39.044  988.62 -14122
## - neighbourhood             29    88.486 1038.06 -13827
## - room_type                  3   198.514 1148.09 -13050
##
## Step:  AIC=-14411.64
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + host_identity_verified + neighbourhood +
##     latitude + longitude + room_type + accommodates + bathrooms +
##     bedrooms + beds + minimum_nights + maximum_nights + minimum_minimum_nights +
##     minimum_maximum_nights + maximum_maximum_nights + minimum_nights_avg_ntm +
##     maximum_nights_avg_ntm + availability_30 + availability_60 +
##     availability_90 + availability_365 + number_of_reviews +
##     number_of_reviews_ltm + number_of_reviews_l30d + instant_bookable +
##     reviews_per_month + HAS + distance_meters + distance_group +
##     bathroom_per_person + bedroom_per_person
##
##                            Df Sum of Sq     RSS    AIC
## - maximum_nights_avg_ntm     1     0.015  949.60 -14414
## - minimum_maximum_nights     1     0.021  949.61 -14414
## - number_of_reviews_ltm      1     0.045  949.63 -14413
## - host_identity_verified     1     0.080  949.66 -14413
## - beds                       1     0.085  949.67 -14413
## - minimum_nights_avg_ntm     1     0.100  949.68 -14413
## - maximum_maximum_nights     1     0.144  949.73 -14413
## - accommodates               1     0.144  949.73 -14412
## - latitude                   1     0.161  949.75 -14412
## - availability_60            1     0.174  949.76 -14412
## - maximum_nights             1     0.207  949.79 -14412
## - longitude                  1     0.250  949.83 -14412
## - host_has_profile_pic       1     0.254  949.84 -14412
## - reviews_per_month          1     0.255  949.84 -14412
```

```
## <none>                                        949.58 -14412
## - minimum_nights             1     0.279  949.86 -14412
## - minimum_minimum_nights     1     0.485  950.07 -14410
## + host_is_superhost          1     0.013  949.57 -14410
## + host_since_years           1     0.010  949.57 -14410
## - host_response_rate         1     0.519  950.10 -14410
## + maximum_minimum_nights     1     0.000  949.58 -14410
## - host_acceptance_rate       1     0.549  950.13 -14410
## - availability_90            1     0.554  950.14 -14409
## - host_verifications         4     1.950  951.53 -14405
## - distance_meters            1     1.885  951.47 -14399
## - host_response_time         3     3.386  952.97 -14392
## - number_of_reviews          1     3.468  953.05 -14387
## - host_total_listings_count  1     4.089  953.67 -14383
## - distance_group             3     5.755  955.34 -14374
## - host_listings_count        1     8.260  957.84 -14351
## - instant_bookable           1     8.350  957.93 -14351
## - number_of_reviews_l30d     1     8.950  958.53 -14346
## - bathroom_per_person        1     9.810  959.39 -14340
## - bedroom_per_person         1    10.852  960.44 -14332
## - bathrooms                  1    15.113  964.70 -14300
## - availability_30            1    15.359  964.94 -14298
## - HAS                        1    17.915  967.50 -14279
## - availability_365           1    19.703  969.29 -14266
## - bedrooms                   1    39.101  988.69 -14123
## - neighbourhood             29    88.641 1038.23 -13828
## - room_type                  3   198.501 1148.09 -13052
##
## Step:  AIC=-14413.53
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + host_identity_verified + neighbourhood +
##     latitude + longitude + room_type + accommodates + bathrooms +
##     bedrooms + beds + minimum_nights + maximum_nights + minimum_minimum_nights +
##     minimum_maximum_nights + maximum_maximum_nights + minimum_nights_avg_ntm +
##     availability_30 + availability_60 + availability_90 + availability_365 +
##     number_of_reviews + number_of_reviews_ltm + number_of_reviews_l30d +
##     instant_bookable + reviews_per_month + HAS + distance_meters +
##     distance_group + bathroom_per_person + bedroom_per_person
##
##                             Df Sum of Sq     RSS    AIC
## - number_of_reviews_ltm      1     0.044  949.64 -14415
## - host_identity_verified     1     0.080  949.68 -14415
## - beds                       1     0.088  949.69 -14415
## - minimum_nights_avg_ntm     1     0.100  949.70 -14415
## - accommodates               1     0.152  949.75 -14414
## - latitude                   1     0.160  949.76 -14414
## - minimum_maximum_nights     1     0.171  949.77 -14414
## - availability_60            1     0.171  949.77 -14414
## - maximum_maximum_nights     1     0.184  949.78 -14414
## - maximum_nights             1     0.203  949.80 -14414
## - longitude                  1     0.250  949.85 -14414
## - host_has_profile_pic       1     0.254  949.85 -14414
## - reviews_per_month          1     0.257  949.86 -14414
```

```
## <none>                                  949.60 -14414
## - minimum_nights              1     0.279  949.88 -14413
## - minimum_minimum_nights      1     0.485  950.09 -14412
## + maximum_nights_avg_ntm      1     0.015  949.58 -14412
## + host_is_superhost           1     0.012  949.59 -14412
## + host_since_years            1     0.011  949.59 -14412
## - host_response_rate          1     0.521  950.12 -14412
## + maximum_minimum_nights      1     0.001  949.60 -14412
## - host_acceptance_rate        1     0.555  950.16 -14411
## - availability_90             1     0.560  950.16 -14411
## - host_verifications          4     1.952  951.55 -14407
## - distance_meters             1     1.883  951.48 -14401
## - host_response_time          3     3.391  952.99 -14394
## - number_of_reviews           1     3.461  953.06 -14389
## - host_total_listings_count   1     4.103  953.70 -14384
## - distance_group              3     5.761  955.36 -14376
## - host_listings_count         1     8.256  957.86 -14353
## - instant_bookable            1     8.366  957.97 -14352
## - number_of_reviews_l30d      1     8.951  958.55 -14348
## - bathroom_per_person         1     9.811  959.41 -14342
## - bedroom_per_person          1    10.878  960.48 -14334
## - bathrooms                   1    15.107  964.71 -14302
## - availability_30             1    15.350  964.95 -14300
## - HAS                         1    17.951  967.55 -14281
## - availability_365            1    19.881  969.48 -14266
## - bedrooms                    1    39.260  988.86 -14124
## - neighbourhood              29    88.680 1038.28 -13829
## - room_type                   3   198.487 1148.09 -13054
##
## Step:  AIC=-14415.19
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + host_identity_verified + neighbourhood +
##     latitude + longitude + room_type + accommodates + bathrooms +
##     bedrooms + beds + minimum_nights + maximum_nights + minimum_minimum_nights +
##     minimum_maximum_nights + maximum_maximum_nights + minimum_nights_avg_ntm +
##     availability_30 + availability_60 + availability_90 + availability_365 +
##     number_of_reviews + number_of_reviews_l30d + instant_bookable +
##     reviews_per_month + HAS + distance_meters + distance_group +
##     bathroom_per_person + bedroom_per_person
##
##                             Df Sum of Sq     RSS    AIC
## - host_identity_verified     1     0.077  949.72 -14417
## - beds                       1     0.088  949.73 -14416
## - minimum_nights_avg_ntm     1     0.098  949.74 -14416
## - accommodates               1     0.150  949.79 -14416
## - latitude                   1     0.160  949.80 -14416
## - availability_60            1     0.172  949.82 -14416
## - minimum_maximum_nights     1     0.174  949.82 -14416
## - maximum_maximum_nights     1     0.190  949.83 -14416
## - maximum_nights             1     0.205  949.85 -14416
## - reviews_per_month          1     0.217  949.86 -14416
## - longitude                  1     0.250  949.89 -14415
## - host_has_profile_pic       1     0.261  949.91 -14415
```

```
## <none>                                    949.64 -14415
## - minimum_nights             1     0.274  949.92 -14415
## + number_of_reviews_ltm      1     0.044  949.60 -14414
## - minimum_minimum_nights     1     0.488  950.13 -14414
## + host_is_superhost          1     0.023  949.62 -14413
## + maximum_nights_avg_ntm     1     0.014  949.63 -14413
## - host_response_rate         1     0.517  950.16 -14413
## + host_since_years           1     0.011  949.63 -14413
## + maximum_minimum_nights     1     0.001  949.64 -14413
## - availability_90            1     0.557  950.20 -14413
## - host_acceptance_rate       1     0.567  950.21 -14413
## - host_verifications         4     1.946  951.59 -14408
## - distance_meters            1     1.884  951.53 -14403
## - host_response_time         3     3.389  953.03 -14396
## - host_total_listings_count  1     4.068  953.71 -14386
## - number_of_reviews          1     4.115  953.76 -14386
## - distance_group             3     5.773  955.42 -14378
## - host_listings_count        1     8.216  957.86 -14355
## - instant_bookable           1     8.373  958.02 -14354
## - number_of_reviews_l30d     1     9.074  958.72 -14349
## - bathroom_per_person        1     9.827  959.47 -14343
## - bedroom_per_person         1    10.850  960.49 -14336
## - bathrooms                  1    15.135  964.78 -14304
## - availability_30            1    15.316  964.96 -14302
## - HAS                        1    17.910  967.55 -14283
## - availability_365           1    19.891  969.53 -14268
## - bedrooms                   1    39.236  988.88 -14126
## - neighbourhood             29    88.709 1038.35 -13831
## - room_type                  3   198.617 1148.26 -13055
##
## Step:  AIC=-14416.61
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + latitude + longitude +
##     room_type + accommodates + bathrooms + bedrooms + beds +
##     minimum_nights + maximum_nights + minimum_minimum_nights +
##     minimum_maximum_nights + maximum_maximum_nights + minimum_nights_avg_ntm +
##     availability_30 + availability_60 + availability_90 + availability_365 +
##     number_of_reviews + number_of_reviews_l30d + instant_bookable +
##     reviews_per_month + HAS + distance_meters + distance_group +
##     bathroom_per_person + bedroom_per_person
##
##                             Df Sum of Sq     RSS    AIC
## - beds                       1     0.082  949.80 -14418
## - minimum_nights_avg_ntm     1     0.096  949.82 -14418
## - accommodates               1     0.146  949.87 -14418
## - latitude                   1     0.165  949.89 -14417
## - minimum_maximum_nights     1     0.172  949.89 -14417
## - availability_60            1     0.174  949.90 -14417
## - maximum_maximum_nights     1     0.187  949.91 -14417
## - maximum_nights             1     0.208  949.93 -14417
## - reviews_per_month          1     0.216  949.94 -14417
## - longitude                  1     0.244  949.97 -14417
## - host_has_profile_pic       1     0.251  949.97 -14417
```

```
## <none>                                   949.72 -14417
## - minimum_nights              1     0.275  950.00 -14416
## + host_identity_verified      1     0.077  949.64 -14415
## - minimum_minimum_nights      1     0.485  950.21 -14415
## + number_of_reviews_ltm       1     0.041  949.68 -14415
## - host_response_rate          1     0.512  950.23 -14415
## + host_is_superhost           1     0.016  949.71 -14415
## + maximum_nights_avg_ntm      1     0.014  949.71 -14415
## + host_since_years            1     0.010  949.71 -14415
## + maximum_minimum_nights      1     0.001  949.72 -14415
## - availability_90             1     0.557  950.28 -14414
## - host_acceptance_rate        1     0.560  950.28 -14414
## - host_verifications          4     1.922  951.64 -14410
## - distance_meters             1     1.877  951.60 -14404
## - host_response_time          3     3.356  953.08 -14397
## - host_total_listings_count   1     4.064  953.79 -14388
## - number_of_reviews           1     4.175  953.90 -14387
## - distance_group              3     5.777  955.50 -14379
## - host_listings_count         1     8.257  957.98 -14356
## - instant_bookable            1     8.359  958.08 -14356
## - number_of_reviews_l30d      1     9.092  958.81 -14350
## - bathroom_per_person         1     9.803  959.52 -14345
## - bedroom_per_person          1    10.852  960.57 -14337
## - bathrooms                   1    15.120  964.84 -14305
## - availability_30             1    15.350  965.07 -14303
## - HAS                         1    17.969  967.69 -14284
## - availability_365            1    19.904  969.63 -14269
## - bedrooms                    1    39.234  988.96 -14128
## - neighbourhood              29    88.960 1038.68 -13831
## - room_type                   3   198.603 1148.32 -13057
##
## Step:  AIC=-14417.98
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + latitude + longitude +
##     room_type + accommodates + bathrooms + bedrooms + minimum_nights +
##     maximum_nights + minimum_minimum_nights + minimum_maximum_nights +
##     maximum_maximum_nights + minimum_nights_avg_ntm + availability_30 +
##     availability_60 + availability_90 + availability_365 + number_of_reviews +
##     number_of_reviews_l30d + instant_bookable + reviews_per_month +
##     HAS + distance_meters + distance_group + bathroom_per_person +
##     bedroom_per_person
##
##                             Df Sum of Sq     RSS    AIC
## - accommodates               1     0.078  949.88 -14419
## - minimum_nights_avg_ntm     1     0.096  949.90 -14419
## - minimum_maximum_nights     1     0.166  949.97 -14419
## - latitude                   1     0.168  949.97 -14419
## - availability_60            1     0.170  949.97 -14419
## - maximum_maximum_nights     1     0.181  949.98 -14419
## - maximum_nights             1     0.199  950.00 -14418
## - reviews_per_month          1     0.233  950.04 -14418
## - longitude                  1     0.248  950.05 -14418
## - host_has_profile_pic       1     0.261  950.06 -14418
```

```
## <none>                                       949.80 -14418
## - minimum_nights            1     0.275  950.08 -14418
## + beds                      1     0.082  949.72 -14417
## + host_identity_verified    1     0.071  949.73 -14416
## - minimum_minimum_nights    1     0.485  950.29 -14416
## + number_of_reviews_ltm     1     0.042  949.76 -14416
## - host_response_rate        1     0.504  950.31 -14416
## + host_is_superhost         1     0.018  949.79 -14416
## + maximum_nights_avg_ntm    1     0.016  949.79 -14416
## + host_since_years          1     0.012  949.79 -14416
## + maximum_minimum_nights    1     0.001  949.80 -14416
## - host_acceptance_rate      1     0.557  950.36 -14416
## - availability_90           1     0.567  950.37 -14416
## - host_verifications        4     1.932  951.74 -14411
## - distance_meters           1     1.868  951.67 -14406
## - host_response_time        3     3.353  953.16 -14399
## - host_total_listings_count  1     4.050  953.85 -14389
## - number_of_reviews         1     4.120  953.92 -14389
## - distance_group            3     5.797  955.60 -14380
## - host_listings_count       1     8.225  958.03 -14358
## - instant_bookable          1     8.295  958.10 -14357
## - number_of_reviews_l30d    1     9.096  958.90 -14351
## - bathroom_per_person       1     9.814  959.62 -14346
## - bedroom_per_person        1    10.770  960.57 -14339
## - bathrooms                 1    15.177  964.98 -14306
## - availability_30           1    15.315  965.12 -14305
## - HAS                       1    17.897  967.70 -14286
## - availability_365          1    19.972  969.78 -14270
## - bedrooms                  1    39.458  989.26 -14127
## - neighbourhood            29    89.015 1038.82 -13832
## - room_type                 3   199.292 1149.10 -13054
##
## Step:  AIC=-14419.39
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + latitude + longitude +
##     room_type + bathrooms + bedrooms + minimum_nights + maximum_nights +
##     minimum_minimum_nights + minimum_maximum_nights + maximum_maximum_nights +
##     minimum_nights_avg_ntm + availability_30 + availability_60 +
##     availability_90 + availability_365 + number_of_reviews +
##     number_of_reviews_l30d + instant_bookable + reviews_per_month +
##     HAS + distance_meters + distance_group + bathroom_per_person +
##     bedroom_per_person
##
##                            Df Sum of Sq     RSS    AIC
## - minimum_nights_avg_ntm    1     0.098  949.98 -14421
## - latitude                  1     0.166  950.05 -14420
## - minimum_maximum_nights    1     0.166  950.05 -14420
## - availability_60           1     0.175  950.06 -14420
## - maximum_maximum_nights    1     0.180  950.06 -14420
## - maximum_nights            1     0.203  950.09 -14420
## - reviews_per_month         1     0.231  950.11 -14420
## - longitude                 1     0.241  950.12 -14420
## - host_has_profile_pic      1     0.257  950.14 -14419
```

```
## <none>                                    949.88 -14419
## - minimum_nights             1     0.274  950.16 -14419
## + accommodates               1     0.078  949.80 -14418
## + host_identity_verified     1     0.071  949.81 -14418
## - minimum_minimum_nights     1     0.485  950.37 -14418
## + number_of_reviews_ltm      1     0.041  949.84 -14418
## + host_is_superhost          1     0.022  949.86 -14418
## + maximum_nights_avg_ntm     1     0.021  949.86 -14418
## - host_response_rate         1     0.513  950.39 -14418
## + beds                       1     0.015  949.87 -14418
## + host_since_years           1     0.013  949.87 -14418
## + maximum_minimum_nights     1     0.001  949.88 -14417
## - host_acceptance_rate       1     0.553  950.43 -14417
## - availability_90            1     0.561  950.44 -14417
## - host_verifications         4     1.933  951.82 -14413
## - distance_meters            1     1.855  951.74 -14407
## - host_response_time         3     3.335  953.22 -14400
## - host_total_listings_count  1     4.032  953.91 -14391
## - number_of_reviews          1     4.146  954.03 -14390
## - distance_group             3     5.800  955.68 -14382
## - host_listings_count        1     8.230  958.11 -14359
## - instant_bookable           1     8.259  958.14 -14359
## - number_of_reviews_l30d     1     9.082  958.96 -14353
## - bathroom_per_person        1     9.741  959.62 -14348
## - bathrooms                  1    15.109  964.99 -14308
## - availability_30            1    15.365  965.25 -14306
## - bedroom_per_person         1    15.470  965.35 -14305
## - HAS                        1    17.953  967.84 -14287
## - availability_365           1    19.940  969.82 -14272
## - neighbourhood             29    89.113 1038.99 -13832
## - bedrooms                   1    96.833 1046.72 -13723
## - room_type                  3   202.522 1152.40 -13035
##
## Step:  AIC=-14420.65
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + latitude + longitude +
##     room_type + bathrooms + bedrooms + minimum_nights + maximum_nights +
##     minimum_minimum_nights + minimum_maximum_nights + maximum_maximum_nights +
##     availability_30 + availability_60 + availability_90 + availability_365 +
##     number_of_reviews + number_of_reviews_l30d + instant_bookable +
##     reviews_per_month + HAS + distance_meters + distance_group +
##     bathroom_per_person + bedroom_per_person
##
##                             Df Sum of Sq     RSS    AIC
## - latitude                   1     0.164  950.14 -14421
## - availability_60            1     0.173  950.15 -14421
## - minimum_maximum_nights     1     0.197  950.18 -14421
## - maximum_nights             1     0.207  950.19 -14421
## - maximum_maximum_nights     1     0.213  950.19 -14421
## - reviews_per_month          1     0.238  950.22 -14421
## - longitude                  1     0.242  950.22 -14421
## - minimum_nights             1     0.242  950.22 -14421
## - host_has_profile_pic       1     0.260  950.24 -14421
```

```
## <none>                                    949.98 -14421
## - minimum_minimum_nights     1     0.401  950.38 -14420
## + minimum_nights_avg_ntm     1     0.098  949.88 -14419
## + maximum_minimum_nights     1     0.088  949.89 -14419
## + accommodates               1     0.080  949.90 -14419
## + host_identity_verified     1     0.069  949.91 -14419
## + number_of_reviews_ltm      1     0.039  949.94 -14419
## - host_response_rate         1     0.507  950.49 -14419
## + host_is_superhost          1     0.021  949.96 -14419
## + maximum_nights_avg_ntm     1     0.021  949.96 -14419
## + beds                       1     0.014  949.97 -14419
## + host_since_years           1     0.014  949.97 -14419
## - host_acceptance_rate       1     0.560  950.54 -14418
## - availability_90            1     0.566  950.55 -14418
## - host_verifications         4     1.960  951.94 -14414
## - distance_meters            1     1.859  951.84 -14409
## - host_response_time         3     3.338  953.32 -14401
## - host_total_listings_count  1     4.068  954.05 -14392
## - number_of_reviews          1     4.150  954.13 -14391
## - distance_group             3     5.778  955.76 -14383
## - instant_bookable           1     8.252  958.23 -14360
## - host_listings_count        1     8.288  958.27 -14360
## - number_of_reviews_l30d     1     9.128  959.11 -14354
## - bathroom_per_person        1     9.721  959.70 -14349
## - bathrooms                  1    15.101  965.08 -14309
## - availability_30            1    15.373  965.35 -14307
## - bedroom_per_person         1    15.479  965.46 -14306
## - HAS                        1    17.933  967.91 -14288
## - availability_365           1    20.115  970.10 -14272
## - neighbourhood             29    89.098 1039.08 -13834
## - bedrooms                   1    96.794 1046.77 -13725
## - room_type                  3   203.236 1153.22 -13032
##
## Step:  AIC=-14421.41
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + longitude + room_type +
##     bathrooms + bedrooms + minimum_nights + maximum_nights +
##     minimum_minimum_nights + minimum_maximum_nights + maximum_maximum_nights +
##     availability_30 + availability_60 + availability_90 + availability_365 +
##     number_of_reviews + number_of_reviews_l30d + instant_bookable +
##     reviews_per_month + HAS + distance_meters + distance_group +
##     bathroom_per_person + bedroom_per_person
##
##                            Df Sum of Sq     RSS    AIC
## - availability_60           1     0.171  950.31 -14422
## - minimum_maximum_nights    1     0.197  950.34 -14422
## - maximum_nights            1     0.203  950.35 -14422
## - maximum_maximum_nights    1     0.212  950.36 -14422
## - minimum_nights            1     0.232  950.38 -14422
## - reviews_per_month         1     0.238  950.38 -14422
## - longitude                 1     0.243  950.39 -14422
## - host_has_profile_pic      1     0.263  950.41 -14421
## <none>                                   950.14 -14421
```

```
## + latitude                   1     0.164  949.98 -14421
## - minimum_minimum_nights     1     0.416  950.56 -14420
## + minimum_nights_avg_ntm     1     0.096  950.05 -14420
## + maximum_minimum_nights     1     0.088  950.06 -14420
## + accommodates               1     0.078  950.07 -14420
## + host_identity_verified     1     0.073  950.07 -14420
## - host_response_rate         1     0.483  950.63 -14420
## + number_of_reviews_ltm      1     0.039  950.10 -14420
## + host_is_superhost          1     0.023  950.12 -14420
## + maximum_nights_avg_ntm     1     0.020  950.12 -14420
## + beds                       1     0.016  950.13 -14420
## + host_since_years           1     0.015  950.13 -14420
## - host_acceptance_rate       1     0.552  950.70 -14419
## - availability_90            1     0.572  950.72 -14419
## - host_verifications         4     1.955  952.10 -14415
## - distance_meters            1     2.091  952.23 -14408
## - host_response_time         3     3.311  953.45 -14402
## - host_total_listings_count  1     4.046  954.19 -14393
## - number_of_reviews          1     4.188  954.33 -14392
## - distance_group             3     5.696  955.84 -14384
## - instant_bookable           1     8.234  958.38 -14361
## - host_listings_count        1     8.262  958.41 -14361
## - number_of_reviews_l30d     1     9.125  959.27 -14355
## - bathroom_per_person        1     9.800  959.94 -14350
## - bathrooms                  1    15.253  965.40 -14309
## - availability_30            1    15.380  965.52 -14308
## - bedroom_per_person         1    15.418  965.56 -14308
## - HAS                        1    17.914  968.06 -14289
## - availability_365           1    20.140  970.28 -14273
## - neighbourhood             29    88.937 1039.08 -13836
## - bedrooms                   1    96.662 1046.81 -13727
## - room_type                  3   203.904 1154.05 -13029
##
## Step:  AIC=-14422.12
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + longitude + room_type +
##     bathrooms + bedrooms + minimum_nights + maximum_nights +
##     minimum_minimum_nights + minimum_maximum_nights + maximum_maximum_nights +
##     availability_30 + availability_90 + availability_365 + number_of_reviews +
##     number_of_reviews_l30d + instant_bookable + reviews_per_month +
##     HAS + distance_meters + distance_group + bathroom_per_person +
##     bedroom_per_person
##
##                            Df Sum of Sq     RSS    AIC
## - maximum_nights            1     0.202  950.52 -14423
## - minimum_maximum_nights    1     0.209  950.52 -14422
## - maximum_maximum_nights    1     0.220  950.53 -14422
## - minimum_nights            1     0.235  950.55 -14422
## - reviews_per_month         1     0.239  950.55 -14422
## - longitude                 1     0.241  950.56 -14422
## - host_has_profile_pic      1     0.260  950.57 -14422
## <none>                                   950.31 -14422
## + availability_60           1     0.171  950.14 -14421
```

```
## + latitude                   1     0.161  950.15 -14421
## - minimum_minimum_nights     1     0.412  950.73 -14421
## + minimum_nights_avg_ntm     1     0.095  950.22 -14421
## + maximum_minimum_nights     1     0.087  950.23 -14421
## + accommodates               1     0.082  950.23 -14421
## + host_identity_verified     1     0.076  950.24 -14421
## + number_of_reviews_ltm      1     0.040  950.27 -14420
## - host_response_rate         1     0.495  950.81 -14420
## + host_is_superhost          1     0.023  950.29 -14420
## + maximum_nights_avg_ntm     1     0.018  950.30 -14420
## + host_since_years           1     0.014  950.30 -14420
## + beds                       1     0.014  950.30 -14420
## - host_acceptance_rate       1     0.547  950.86 -14420
## - host_verifications         4     1.966  952.28 -14415
## - distance_meters            1     2.088  952.40 -14408
## - host_response_time         3     3.315  953.63 -14403
## - host_total_listings_count  1     4.066  954.38 -14393
## - number_of_reviews          1     4.190  954.50 -14392
## - distance_group             3     5.697  956.01 -14385
## - availability_90            1     6.280  956.59 -14377
## - instant_bookable           1     8.252  958.57 -14362
## - host_listings_count        1     8.286  958.60 -14362
## - number_of_reviews_l30d     1     9.089  959.40 -14356
## - bathroom_per_person        1     9.765  960.08 -14351
## - bathrooms                  1    15.214  965.53 -14310
## - bedroom_per_person         1    15.348  965.66 -14309
## - HAS                        1    17.919  968.23 -14290
## - availability_365           1    20.630  970.94 -14270
## - availability_30            1    29.257  979.57 -14206
## - neighbourhood             29    89.019 1039.33 -13836
## - bedrooms                   1    96.647 1046.96 -13728
## - room_type                  3   204.157 1154.47 -13028
##
## Step:  AIC=-14422.59
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + longitude + room_type +
##     bathrooms + bedrooms + minimum_nights + minimum_minimum_nights +
##     minimum_maximum_nights + maximum_maximum_nights + availability_30 +
##     availability_90 + availability_365 + number_of_reviews +
##     number_of_reviews_l30d + instant_bookable + reviews_per_month +
##     HAS + distance_meters + distance_group + bathroom_per_person +
##     bedroom_per_person
##
##                            Df Sum of Sq     RSS    AIC
## - maximum_maximum_nights     1     0.156  950.67 -14423
## - minimum_maximum_nights     1     0.184  950.70 -14423
## - reviews_per_month          1     0.204  950.72 -14423
## - minimum_nights             1     0.232  950.75 -14423
## - longitude                  1     0.244  950.76 -14423
## - host_has_profile_pic       1     0.253  950.77 -14423
## <none>                                   950.52 -14423
## + maximum_nights             1     0.202  950.31 -14422
## + availability_60            1     0.170  950.35 -14422
```

```
## + latitude                   1     0.157  950.36 -14422
## - minimum_minimum_nights     1     0.418  950.93 -14421
## + minimum_nights_avg_ntm     1     0.098  950.42 -14421
## + maximum_minimum_nights     1     0.090  950.43 -14421
## + accommodates               1     0.086  950.43 -14421
## + host_identity_verified     1     0.079  950.44 -14421
## - host_response_rate         1     0.483  951.00 -14421
## + number_of_reviews_ltm      1     0.042  950.47 -14421
## + host_is_superhost          1     0.024  950.49 -14421
## + maximum_nights_avg_ntm     1     0.013  950.50 -14421
## + host_since_years           1     0.010  950.51 -14421
## + beds                       1     0.010  950.51 -14421
## - host_acceptance_rate       1     0.560  951.08 -14420
## - host_verifications         4     1.917  952.43 -14416
## - distance_meters            1     2.092  952.61 -14409
## - host_response_time         3     3.331  953.85 -14403
## - host_total_listings_count  1     4.074  954.59 -14394
## - number_of_reviews          1     4.630  955.15 -14390
## - distance_group             3     5.713  956.23 -14386
## - availability_90            1     6.202  956.72 -14378
## - instant_bookable           1     8.217  958.73 -14363
## - host_listings_count        1     8.283  958.80 -14362
## - number_of_reviews_l30d     1     9.050  959.57 -14356
## - bathroom_per_person        1     9.800  960.32 -14351
## - bedroom_per_person         1    15.242  965.76 -14310
## - bathrooms                  1    15.300  965.82 -14310
## - HAS                        1    18.172  968.69 -14288
## - availability_365           1    20.450  970.97 -14272
## - availability_30            1    29.214  979.73 -14207
## - neighbourhood             29    88.949 1039.47 -13837
## - bedrooms                   1    96.468 1046.98 -13729
## - room_type                  3   204.067 1154.58 -13030
##
## Step:  AIC=-14423.41
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + longitude + room_type +
##     bathrooms + bedrooms + minimum_nights + minimum_minimum_nights +
##     minimum_maximum_nights + availability_30 + availability_90 +
##     availability_365 + number_of_reviews + number_of_reviews_l30d +
##     instant_bookable + reviews_per_month + HAS + distance_meters +
##     distance_group + bathroom_per_person + bedroom_per_person
##
##                            Df Sum of Sq     RSS    AIC
## - minimum_maximum_nights     1     0.033  950.70 -14425
## - reviews_per_month          1     0.205  950.88 -14424
## - minimum_nights             1     0.232  950.90 -14424
## - longitude                  1     0.234  950.91 -14424
## - host_has_profile_pic       1     0.261  950.93 -14423
## <none>                                   950.67 -14423
## + availability_60            1     0.177  950.50 -14423
## + latitude                   1     0.157  950.52 -14423
## + maximum_maximum_nights     1     0.156  950.52 -14423
## + maximum_nights             1     0.138  950.53 -14422
```

```
## + minimum_nights_avg_ntm     1     0.125  950.55 -14422
## + maximum_minimum_nights     1     0.117  950.55 -14422
## - minimum_minimum_nights     1     0.423  951.09 -14422
## + accommodates               1     0.086  950.59 -14422
## + host_identity_verified     1     0.076  950.60 -14422
## + maximum_nights_avg_ntm     1     0.048  950.62 -14422
## - host_response_rate         1     0.481  951.15 -14422
## + number_of_reviews_ltm      1     0.047  950.62 -14422
## + host_is_superhost          1     0.026  950.65 -14422
## + host_since_years           1     0.017  950.65 -14422
## + beds                       1     0.008  950.66 -14422
## - host_acceptance_rate       1     0.596  951.27 -14421
## - host_verifications         4     1.929  952.60 -14417
## - distance_meters            1     2.066  952.74 -14410
## - host_response_time         3     3.374  954.05 -14404
## - host_total_listings_count  1     4.062  954.73 -14395
## - number_of_reviews          1     4.608  955.28 -14391
## - distance_group             3     5.725  956.40 -14386
## - availability_90            1     6.263  956.93 -14378
## - host_listings_count        1     8.208  958.88 -14364
## - instant_bookable           1     8.336  959.01 -14363
## - number_of_reviews_l30d     1     9.070  959.74 -14357
## - bathroom_per_person        1     9.841  960.51 -14351
## - bedroom_per_person         1    15.242  965.91 -14311
## - bathrooms                  1    15.316  965.99 -14310
## - HAS                        1    18.058  968.73 -14290
## - availability_365           1    20.765  971.44 -14270
## - availability_30            1    29.241  979.91 -14208
## - neighbourhood             29    89.734 1040.41 -13833
## - bedrooms                   1    96.612 1047.28 -13729
## - room_type                  3   203.932 1154.60 -13032
##
## Step:  AIC=-14425.16
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + longitude + room_type +
##     bathrooms + bedrooms + minimum_nights + minimum_minimum_nights +
##     availability_30 + availability_90 + availability_365 + number_of_reviews +
##     number_of_reviews_l30d + instant_bookable + reviews_per_month +
##     HAS + distance_meters + distance_group + bathroom_per_person +
##     bedroom_per_person
##
##                             Df Sum of Sq     RSS    AIC
## - reviews_per_month          1     0.197  950.90 -14426
## - minimum_nights             1     0.232  950.94 -14425
## - longitude                  1     0.236  950.94 -14425
## - host_has_profile_pic       1     0.256  950.96 -14425
## <none>                                    950.70 -14425
## + availability_60            1     0.181  950.52 -14424
## + maximum_nights             1     0.169  950.54 -14424
## + latitude                   1     0.157  950.55 -14424
## + minimum_nights_avg_ntm     1     0.127  950.58 -14424
## + maximum_minimum_nights     1     0.120  950.58 -14424
## - minimum_minimum_nights     1     0.423  951.13 -14424
```

```
## + accommodates              1     0.087  950.62 -14424
## + host_identity_verified    1     0.078  950.63 -14424
## - host_response_rate        1     0.479  951.18 -14424
## + number_of_reviews_ltm     1     0.042  950.66 -14424
## + minimum_maximum_nights    1     0.033  950.67 -14423
## + host_is_superhost         1     0.022  950.68 -14423
## + maximum_nights_avg_ntm    1     0.020  950.68 -14423
## + host_since_years          1     0.016  950.69 -14423
## + beds                      1     0.008  950.70 -14423
## + maximum_maximum_nights    1     0.004  950.70 -14423
## - host_acceptance_rate      1     0.584  951.29 -14423
## - host_verifications        4     1.914  952.62 -14419
## - distance_meters           1     2.063  952.77 -14412
## - host_response_time        3     3.354  954.06 -14406
## - host_total_listings_count  1     4.076  954.78 -14396
## - number_of_reviews         1     4.833  955.54 -14391
## - distance_group            3     5.734  956.44 -14388
## - availability_90           1     6.262  956.97 -14380
## - host_listings_count       1     8.213  958.92 -14365
## - instant_bookable          1     8.308  959.01 -14365
## - number_of_reviews_l30d    1     9.107  959.81 -14359
## - bathroom_per_person       1     9.850  960.55 -14353
## - bedroom_per_person        1    15.217  965.92 -14313
## - bathrooms                 1    15.315  966.02 -14312
## - HAS                       1    18.109  968.81 -14292
## - availability_365          1    20.733  971.44 -14272
## - availability_30           1    29.329  980.03 -14209
## - neighbourhood            29    89.711 1040.42 -13835
## - bedrooms                  1    96.586 1047.29 -13731
## - room_type                 3   203.913 1154.62 -13034
##
## Step:  AIC=-14425.67
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + longitude + room_type +
##     bathrooms + bedrooms + minimum_nights + minimum_minimum_nights +
##     availability_30 + availability_90 + availability_365 + number_of_reviews +
##     number_of_reviews_l30d + instant_bookable + HAS + distance_meters +
##     distance_group + bathroom_per_person + bedroom_per_person
##
##                            Df Sum of Sq     RSS    AIC
## - minimum_nights            1     0.229  951.13 -14426
## - longitude                 1     0.242  951.14 -14426
## - host_has_profile_pic      1     0.260  951.16 -14426
## <none>                                   950.90 -14426
## + reviews_per_month         1     0.197  950.70 -14425
## + availability_60           1     0.182  950.72 -14425
## + latitude                  1     0.157  950.74 -14425
## + minimum_nights_avg_ntm    1     0.134  950.77 -14425
## + maximum_nights            1     0.133  950.77 -14425
## + maximum_minimum_nights    1     0.126  950.78 -14425
## - minimum_minimum_nights    1     0.410  951.31 -14425
## + accommodates              1     0.085  950.82 -14424
## + host_identity_verified    1     0.076  950.83 -14424
```

```
## - host_response_rate         1     0.473  951.37 -14424
## + host_since_years           1     0.043  950.86 -14424
## + minimum_maximum_nights     1     0.024  950.88 -14424
## + host_is_superhost          1     0.020  950.88 -14424
## + maximum_nights_avg_ntm     1     0.014  950.89 -14424
## + beds                       1     0.013  950.89 -14424
## + number_of_reviews_ltm      1     0.003  950.90 -14424
## + maximum_maximum_nights     1     0.002  950.90 -14424
## - host_acceptance_rate       1     0.546  951.45 -14424
## - host_verifications         4     2.071  952.97 -14418
## - distance_meters            1     2.095  953.00 -14412
## - host_response_time         3     3.299  954.20 -14407
## - host_total_listings_count  1     4.112  955.01 -14397
## - distance_group             3     5.712  956.61 -14389
## - number_of_reviews          1     6.317  957.22 -14380
## - availability_90            1     6.370  957.27 -14380
## - instant_bookable           1     8.176  959.08 -14366
## - host_listings_count        1     8.276  959.18 -14365
## - bathroom_per_person        1     9.796  960.70 -14354
## - bedroom_per_person         1    15.227  966.13 -14313
## - bathrooms                  1    15.254  966.16 -14313
## - number_of_reviews_l30d     1    17.911  968.81 -14294
## - HAS                        1    18.118  969.02 -14292
## - availability_365           1    21.351  972.25 -14268
## - availability_30            1    29.554  980.46 -14208
## - neighbourhood             29    90.136 1041.04 -13832
## - bedrooms                   1    96.713 1047.61 -13731
## - room_type                  3   203.772 1154.67 -13035
##
## Step:  AIC=-14425.94
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + longitude + room_type +
##     bathrooms + bedrooms + minimum_minimum_nights + availability_30 +
##     availability_90 + availability_365 + number_of_reviews +
##     number_of_reviews_l30d + instant_bookable + HAS + distance_meters +
##     distance_group + bathroom_per_person + bedroom_per_person
##
##                            Df Sum of Sq     RSS    AIC
## - longitude                 1     0.246  951.38 -14426
## - host_has_profile_pic      1     0.258  951.39 -14426
## <none>                                   951.13 -14426
## + minimum_nights            1     0.229  950.90 -14426
## + reviews_per_month         1     0.194  950.94 -14425
## + availability_60           1     0.185  950.95 -14425
## + latitude                  1     0.147  950.98 -14425
## + maximum_nights            1     0.132  951.00 -14425
## + maximum_minimum_nights    1     0.113  951.02 -14425
## + minimum_nights_avg_ntm    1     0.097  951.03 -14425
## + accommodates              1     0.084  951.05 -14425
## + host_identity_verified    1     0.077  951.05 -14424
## - host_response_rate        1     0.465  951.60 -14424
## + host_since_years          1     0.041  951.09 -14424
## + minimum_maximum_nights    1     0.025  951.11 -14424
```

```
## + host_is_superhost          1     0.020  951.11 -14424
## + maximum_nights_avg_ntm     1     0.015  951.12 -14424
## + beds                       1     0.013  951.12 -14424
## + number_of_reviews_ltm      1     0.004  951.13 -14424
## + maximum_maximum_nights     1     0.002  951.13 -14424
## - host_acceptance_rate       1     0.537  951.67 -14424
## - host_verifications         4     2.051  953.18 -14418
## - distance_meters            1     2.066  953.20 -14412
## - host_response_time         3     3.326  954.46 -14407
## - minimum_minimum_nights     1     3.134  954.26 -14404
## - host_total_listings_count  1     4.104  955.23 -14397
## - distance_group             3     5.751  956.88 -14389
## - number_of_reviews          1     6.345  957.48 -14380
## - availability_90            1     6.353  957.48 -14380
## - instant_bookable           1     8.136  959.27 -14367
## - host_listings_count        1     8.247  959.38 -14366
## - bathroom_per_person        1     9.838  960.97 -14354
## - bedroom_per_person         1    15.217  966.35 -14314
## - bathrooms                  1    15.276  966.41 -14313
## - number_of_reviews_l30d     1    17.872  969.00 -14294
## - HAS                        1    18.078  969.21 -14292
## - availability_365           1    21.302  972.43 -14269
## - availability_30            1    29.615  980.75 -14207
## - neighbourhood             29    89.999 1041.13 -13834
## - bedrooms                   1    96.657 1047.79 -13732
## - room_type                  3   203.549 1154.68 -13037
##
## Step:  AIC=-14426.08
## LogPrice ~ host_response_time + host_response_rate + host_acceptance_rate +
##     host_listings_count + host_total_listings_count + host_verifications +
##     host_has_profile_pic + neighbourhood + room_type + bathrooms +
##     bedrooms + minimum_minimum_nights + availability_30 + availability_90 +
##     availability_365 + number_of_reviews + number_of_reviews_l30d +
##     instant_bookable + HAS + distance_meters + distance_group +
##     bathroom_per_person + bedroom_per_person
##
##                            Df Sum of Sq     RSS    AIC
## <none>                                   951.38 -14426
## - host_has_profile_pic       1     0.265  951.64 -14426
## + longitude                  1     0.246  951.13 -14426
## + minimum_nights             1     0.233  951.14 -14426
## + reviews_per_month          1     0.200  951.18 -14426
## + availability_60            1     0.183  951.19 -14426
## + latitude                   1     0.148  951.23 -14425
## + maximum_nights             1     0.137  951.24 -14425
## + maximum_minimum_nights     1     0.110  951.27 -14425
## + minimum_nights_avg_ntm     1     0.097  951.28 -14425
## + accommodates               1     0.077  951.30 -14425
## + host_identity_verified     1     0.071  951.31 -14425
## - host_response_rate         1     0.476  951.85 -14424
## + host_since_years           1     0.044  951.33 -14424
## + minimum_maximum_nights     1     0.027  951.35 -14424
## + host_is_superhost          1     0.019  951.36 -14424
## + maximum_nights_avg_ntm     1     0.016  951.36 -14424
```

```
## + beds                        1     0.016  951.36 -14424
## + number_of_reviews_ltm       1     0.004  951.37 -14424
## + maximum_maximum_nights      1     0.003  951.37 -14424
## - host_acceptance_rate        1     0.552  951.93 -14424
## - host_verifications          4     2.038  953.42 -14419
## - distance_meters             1     2.225  953.60 -14411
## - host_response_time          3     3.324  954.70 -14407
## - minimum_minimum_nights      1     3.116  954.49 -14405
## - host_total_listings_count   1     4.084  955.46 -14397
## - distance_group              3     6.410  957.79 -14384
## - number_of_reviews           1     6.328  957.71 -14380
## - availability_90             1     6.410  957.79 -14380
## - instant_bookable            1     8.139  959.52 -14367
## - host_listings_count         1     8.184  959.56 -14366
## - bathroom_per_person         1     9.814  961.19 -14354
## - bathrooms                   1    15.214  966.59 -14314
## - bedroom_per_person          1    15.227  966.60 -14314
## - number_of_reviews_l30d      1    17.936  969.31 -14294
## - HAS                         1    17.999  969.38 -14293
## - availability_365            1    21.270  972.65 -14269
## - availability_30             1    29.733  981.11 -14207
## - bedrooms                    1    96.898 1048.27 -13730
## - neighbourhood              29   123.344 1074.72 -13607
## - room_type                   3   203.345 1154.72 -13039
```

```r
# Predict on test data (you should also remove 'price' from test2)
test2_noprice <- test2[, !names(test2) %in% "price"]
pred2_s <- predict(step_model2, newdata = test2_noprice)

# Convert log(price + 1) back to original price
pred2_S <- exp(pred2_s) - 1

# Validate and filter for finite predictions
valid_idx <- which(!is.na(pred2_S) & !is.infinite(pred2_S))
actual_price_valid <- actual_price[valid_idx]
pred2_S_valid <- pred2_S[valid_idx]

# Evaluate performance
test_mse2_S <- mean((actual_price_valid - pred2_S_valid)^2)
rmse_S <- sqrt(test_mse2_S)
r2_step <- 1 - sum((pred2_S_valid - actual_price_valid)^2) /
  sum((actual_price_valid - mean(actual_price_valid))^2)

# Output
cat("Test MSE for Stepwise:", round(test_mse2_S, 2), "\n")
```

```
## Test MSE for Stepwise: 10490.03
```

```r
cat("Test RMSE for Stepwise:", round(rmse_S, 2), "\n")
```

```
## Test RMSE for Stepwise: 102.42
```

```r
cat("Test R^2 for Stepwise:", round(r2_step, 4), "\n")
```

```
## Test R^2 for Stepwise: 0.5365
```

Random forest selection

```r
library(randomForest)
```

```
## Warning: package 'randomForest' was built under R version 4.3.3

## randomForest 4.7-1.2

## Type rfNews() to see new features/changes/bug fixes.

##
## Attaching package: 'randomForest'

## The following object is masked from 'package:dplyr':
##
##     combine
```

```r
set.seed(123)
rf_model <- randomForest(x = as.data.frame(x_train2),
                         y = y_train2,
                         ntree = 500,
                         mtry = floor(sqrt(ncol(x_train2))),
                         importance = TRUE)
y_pred_rf <- predict(rf_model, newdata = as.data.frame(x_test2))
#logprice->price
price_pred_rf <- exp(y_pred_rf) - 1
actual_price <- test2$price

test_mse_rf <- mean((actual_price - price_pred_rf)^2)
rmse_rf <- sqrt(test_mse_rf)

cat("Test MSE for random forest: ", test_mse_rf, "\n")
```

```
## Test MSE for random forest:  8626.707
```

```r
cat("Test RMSE for random forest: ", rmse_rf, "\n")
```

```
## Test RMSE for random forest:  92.88007
```

```r
# if random forest win
#importance(rf_model)
#varImpPlot(rf_model)
```

Ridge

```r
library(glmnet)

# Standardize the training and test sets
x_train_std <- scale(x_train2)
x_test_std <- scale(x_test2,
                    center = attr(x_train_std, "scaled:center"),
                    scale = attr(x_train_std, "scaled:scale"))

# Use cross-validation to find the best lambda
set.seed(123)
cv_ridge <- cv.glmnet(x_train_std, y_train2, alpha = 0)  # alpha = 0 for Ridge

best_lambda_ridge <- cv_ridge$lambda.min
cat("Best lambda for Ridge:", best_lambda_ridge, "\n")
```

```
## Best lambda for Ridge: 0.03868529
```

```r
# Fit the Ridge model with the selected lambda
ridge_model <- glmnet(x_train_std, y_train2, alpha = 0, lambda = best_lambda_ridge)

# Predict log(price + 1) on the test set
pred_ridge <- predict(ridge_model, newx = x_test_std)

# Limit extreme predicted values to avoid overflow when applying exp()
upper_bound <- quantile(y_train2, 0.995)
lower_bound <- quantile(y_train2, 0.005)
pred_ridge[pred_ridge > upper_bound] <- upper_bound
pred_ridge[pred_ridge < lower_bound] <- lower_bound

# Convert predicted log(price + 1) back to original price
pred_R <- exp(pred_ridge) - 1
actual_price <- test2$price

# Compute MSE and RMSE in original price scale
mse_ridge <- mean((actual_price - pred_R)^2, na.rm = TRUE)
rmse_R <- sqrt(mse_ridge)

cat("Test MSE for Ridge:", mse_ridge, "\n")
```

```
## Test MSE for Ridge: 10419.71
```

```r
cat("Test RMSE for Ridge:", rmse_R, "\n")
```

```
## Test RMSE for Ridge: 102.077
```

XGBoost

```r
library(xgboost)
```

```
## Warning: package 'xgboost' was built under R version 4.3.3

##
## Attaching package: 'xgboost'

## The following object is masked from 'package:dplyr':
##
##     slice
```

```r
library(Matrix)
library(Metrics)
```

```
## Warning: package 'Metrics' was built under R version 4.3.3

##
## Attaching package: 'Metrics'

## The following object is masked from 'package:pROC':
##
##     auc

## The following object is masked from 'package:rlang':
##
##     ll
```

```r
# Convert data to xgb.DMatrix format
dtrain <- xgb.DMatrix(data = as.matrix(x_train2), label = y_train2)
dtest <- xgb.DMatrix(data = as.matrix(x_test2), label = test2$LogPrice)

# Set parameters
params <- list(
  objective = "reg:squarederror",
  eval_metric = "rmse",
  max_depth = 6,
  eta = 0.1,
  subsample = 0.8,
  colsample_bytree = 0.8
)

# Train model
set.seed(123)
xgb_model <- xgb.train(
  params = params,
  data = dtrain,
  nrounds = 200,
  watchlist = list(train = dtrain, test = dtest),
  early_stopping_rounds = 10,
  verbose = 0
)

# Predict
```

```r
pred_xgb_log <- predict(xgb_model, newdata = dtest)
pred_xgb_price <- exp(pred_xgb_log) - 1
actual_price <- test2$price

# Metrics
mse_xgb <- mean((actual_price - pred_xgb_price)^2)
rmse_xgb <- sqrt(mse_xgb)
r2_xgb <- 1 - sum((actual_price - pred_xgb_price)^2) / sum((actual_price - mean(actual_price))^2)

# Output
cat("Test MSE for XGBoost: ", round(mse_xgb, 2), "\n")
```

```
## Test MSE for XGBoost:  6905.83
```

```r
cat("Test RMSE for XGBoost: ", round(rmse_xgb, 2), "\n")
```

```
## Test RMSE for XGBoost:  83.1
```

```r
cat("Test R2 for XGBoost: ", round(r2_xgb, 3), "\n")
```

```
## Test R2 for XGBoost:  0.695
```

Compare MSE/R^2

```r
#test r^2
r2_lasso <- 1 - sum((pred2_L - actual_price)^2) / sum((actual_price - mean(actual_price))^2)
r2_rf <- 1 - sum((price_pred_rf - actual_price)^2) / sum((actual_price - mean(actual_price))^2)
r2_ridge <- 1 - sum((pred_R - actual_price)^2) / sum((actual_price - mean(actual_price))^2)
r2_step <- 1 - sum((pred2_S_valid - actual_price_valid)^2) /
  sum((actual_price_valid - mean(actual_price_valid))^2)

#train r^2
#lasso
train2_lasso<-predict(lasso_cv2, s = best_lambda2, newx = x_train2)
train2_L<-exp(train2_lasso) - 1
actual_price_train <- train2$price
r2_train_lasso <- 1 - sum((train2_L - actual_price_train)^2) /
                     sum((actual_price_train - mean(actual_price_train))^2)
#ridge
train2_ridge <- predict(ridge_model, newx = x_train_std)
train2_ridge[train2_ridge > upper_bound] <- upper_bound
train2_ridge[train2_ridge < lower_bound] <- lower_bound
train2_R <- exp(train2_ridge) - 1

r2_train_ridge <- 1 - sum((train2_R - actual_price_train)^2) /
                      sum((actual_price_train - mean(actual_price_train))^2)
cat("Ridge Train R²:", round(r2_train_ridge, 4), "\n")
```

```
## Ridge Train R²: 0.5146
```

```r
#rf
train2_rf <- predict(rf_model, newdata = x_train2)
train2_RF <- exp(train2_rf) - 1

r2_train_rf <- 1 - sum((train2_RF - actual_price_train)^2) /
                   sum((actual_price_train - mean(actual_price_train))^2)
cat("Random Forest Train R²:", round(r2_train_rf, 4), "\n")
```

```
## Random Forest Train R²: 0.8885
```

```r
#step
train2_step <- predict(step_model2, newdata = train2_noprice)
train2_S <- exp(train2_step) - 1

r2_train_step <- 1 - sum((train2_S - actual_price_train)^2) /
                     sum((actual_price_train - mean(actual_price_train))^2)
cat("Stepwise Train R²:", round(r2_train_step, 4), "\n")
```

```
## Stepwise Train R²: 0.5153
```

```r
#xgb
# Predict on training set
train_pred_xgb <- predict(xgb_model, newdata = x_train2)

# Convert log(price + 1) back to original price
train_price_pred_xgb <- exp(train_pred_xgb) - 1
actual_price_train <- train2$price

# Calculate Train MSE
train_mse_xgb <- mean((train_price_pred_xgb - actual_price_train)^2)

# Calculate Train RMSE
train_rmse_xgb <- sqrt(train_mse_xgb)

# Train R-squared
r2_train_xgb <- 1 - sum((train_price_pred_xgb - actual_price_train)^2) /
                      sum((actual_price_train - mean(actual_price_train))^2)

cat("XGBoost Train MSE:", round(train_mse_xgb, 2), "\n")
```

```
## XGBoost Train MSE: 2833.15
```

```r
cat("XGBoost Train RMSE:", round(train_rmse_xgb, 2), "\n")
```

```
## XGBoost Train RMSE: 53.23
```

```r
cat("XGBoost Train R²:", round(r2_train_xgb, 4), "\n")
```

```
## XGBoost Train R²: 0.8921
```

```r
#total factors
model_performance <- data.frame(
  Model = c("LASSO", "Linear Regression", "Random Forest", "Ridge", "XGBoost"),
  Test_MSE = c(test_mse2_L, test_mse2_S, test_mse_rf, mse_ridge, mse_xgb),
  Test_RMSE = c(rmse_L, rmse_S, rmse_rf, rmse_R, rmse_xgb),
  Test_R2 = c(r2_lasso, r2_step, r2_rf, r2_ridge, r2_xgb),
  Train_R2 = c(r2_train_lasso, r2_train_step, r2_train_rf, r2_train_ridge, r2_train_xgb)
)

print(model_performance)
```

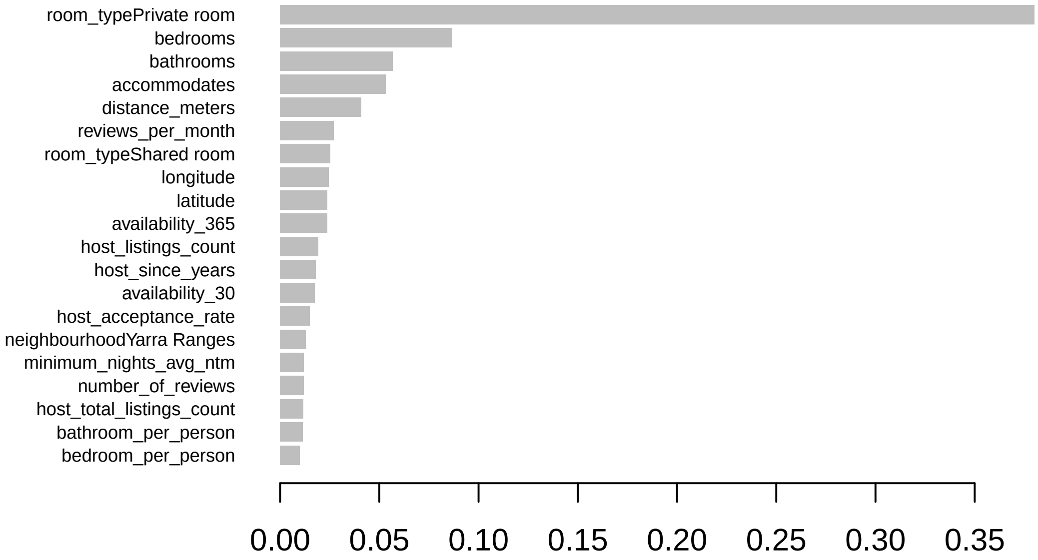
```
##               Model  Test_MSE Test_RMSE   Test_R2  Train_R2
## 1             LASSO 10464.427 102.29578 0.5376244 0.5132455
## 2 Linear Regression 10490.025 102.42082 0.5364934 0.5153254
## 3     Random Forest  8626.707  92.88007 0.6188250 0.8885304
## 4             Ridge 10419.708 102.07697 0.5396004 0.5146293
## 5           XGBoost  6905.832  83.10134 0.6948626 0.8921472
```

```r
summary(test2$price)
```

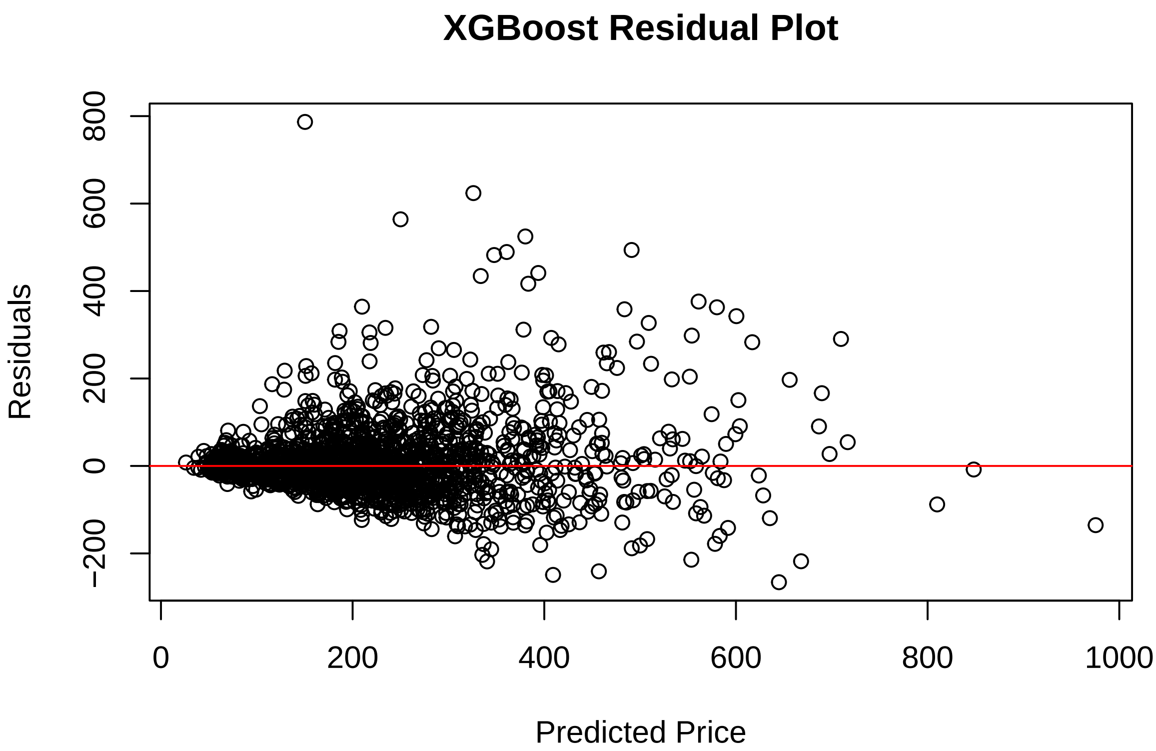
```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    28.0   130.0   202.5   234.3   298.0  1000.0
```

Winner is XGBoost

```r
#important variables
importance_matrix <- xgb.importance(model = xgb_model)
xgb.plot.importance(importance_matrix, top_n = 20)
```




```r
#residual plot
resid_xgb <- actual_price - pred_xgb_price
plot(pred_xgb_price, resid_xgb, main="XGBoost Residual Plot",
     xlab="Predicted Price", ylab="Residuals")
abline(h=0, col="red")
```



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.